

FC7xxx GPT User Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.3	2023/10/20	Updated for MCAL V0.3.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 GPT Introduction

1.1 Requirement Tracing

The design of this module follows the specifications of the GPT driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_GPTDriver*.

1.2 Hardware Summary

The GPT driver provides logical channels which are configured by hardware including Flexible Timer Unit (FTU), Always-on Timer (AONTIMER), Real-Time Clock (RTC), FC Programmable Interrupt Timer (FCPIT) and Timer Stamp (TSTMP).

The GPT driver provides several services including initializing channels, de-initializing channels, getting the elapsed time, getting the time remaining, and others.

The GPT driver implements a maximum of 118 channels on FC7300 peripherals:

- 1) 96 channels are implemented on 12 FTU modules.
- 2) 1 channel is implemented on the AONTIMER module.
- 3) 1 channel is implemented on the RTC module.
- 4) 4 channel is implemented on the FCPIT module.
- 5) 16 channel is implemented on the TSTMP module.

The GPT driver implements a maximum of 78 channels on FC7240 peripherals:

- 1) 96 channels are implemented on 8 FTU modules.
- 2) 1 channel is implemented on the AONTIMER module.
- 3) 1 channel is implemented on the RTC module.
- 4) 4 channel is implemented on the FCPIT module.
- 5) 8 channel is implemented on the TSTMP module.

FTU Channel

A maximum of eight 16-bit counters on each module FTU (0~11).(FC7300)

FTU1 and FTU2 support 24-bit counter width, and other FTUs support 16-bit counter width.(FC7240)

AONTIMER Channel

Independent timeout periods for each 16-bit timer.

RTC Channel

Independent timeout periods for each 32-bit timer.

FCPIT Channel (Only available on FC7240)

32-bit programmable timeout value for each channel.

TSTMP Channel

4 independent 32-bit width modulate timers with individual interrupts supported.

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 1 **SWS_Gpt_00261**

Description	Gpt_Irq.c shall include Gpt.h for the prototype declaration of the notification functions.
Rejection Reason	Considering lots of local variables are defined in Gpt.c, the notification function: process common interrupt is included in Gpt.c.

Rejected Requirement 2 **SWS_Gpt_00337**

Description	Module Dem EcuM Std_Types	Imported Type Dem_EventIdType Dem_EventStatusType EcuM_WakeupSourceType Std_ReturnType Std_VersionInfoType
Rejection Reason	These development errors are not needed	

Rejected Requirement 3 **SWS_Gpt_00353**

Description	If the register can affect several hardware modules and if it is an I/O register it shall be initialized by the PORT driver.
Rejection Reason	Not applicable to FC7XXX series chip

Rejected Requirement 4 **SWS_Gpt_00354**

Description	If the register can affect several hardware modules and if it is not an I/O register it shall be initialized by the MCU driver.
Rejection Reason	Not applicable to FC7XXX series chip

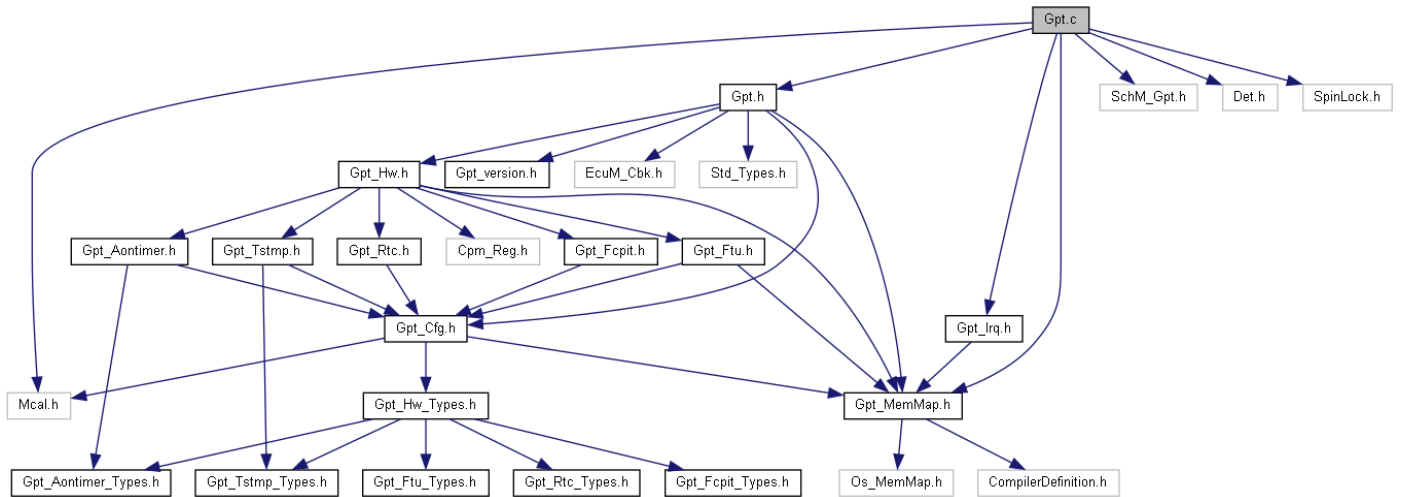
Rejected Requirement 5 **SWS_Gpt_00355**

Description	One-time writable registers that require initialization directly after reset shall be initialized by the startup code.
Rejection Reason	Not applicable to FC7XXX series chip

Rejected Requirement 6 **SWS_Gpt_00356**

Description	All other registers shall be initialized by the startup code.
Rejection Reason	Not applicable to the GPT module

2.2 File Structure



2.3 Macros

2.3.1 Macros in Gpt.h

- **#define GPT_E_INVALID_CALL ((uint8)0xA0U)**
Function Gpt_StartTimer is called when the driver is in sleep mode for a channel which is not wakeup enabled.
- **#define GPT_E_UNINIT ((uint8)0x0AU)**
Function called without module initialization.
- **#define GPT_E_BUSY ((uint8)0x0BU)**
Function called when timer channel is still running.
- **#define GPT_E_ALREADY_INITIALIZED ((uint8)0x0DU)**
Initialization called when already initialized.
- **#define GPT_E_PARAM_CHANNEL ((uint8)0x14U)**
Function called for invalid channel.
- **#define GPT_E_PARAM_VALUE ((uint8)0x15U)**
Function called with parameter value out of range.
- **#define GPT_E_PARAM_POINTER ((uint8)0x16U)**
Function called with NULL pointer.
- **#define GPT_E_MODE ((uint8)0x0C)**
Function called with invalid the parameter in function Gpt_GetPredefTimerValue.
- **#define GPT_E_PARAM_PREDEF_TIMER ((uint8)0x17)**
Function called with invalid the parameter in function Gpt_GetPredefTimerValue.

- **#define GPT_E_INIT_FAILED ((uint8)0x0E)**
Function called with invalid the parameter in function Gpt_Init.
- **#define GPT_E_PARAM_MODE ((uint8)0x1FU)**
Function called with invalid mode param.
- **#define GPT_E_PARAM_PARTITION ((uint8)0x20U)**
Function called with invalid channel for the partition.
- **#define GPT_GETVERSIONINFO_ID ((uint8)0x00U)**
API service ID for Gpt_GetVersionInfo function.
- **#define GPT_INIT_ID ((uint8)0x01U)**
API service ID for Gpt_Init function.
- **#define GPT_DEINIT_ID ((uint8)0x02U)**
API service ID for Gpt_DeInit function.
- **#define GPT_TIMEELAPSED_ID ((uint8)0x03U)**
API service ID for Gpt_GetTimeElapsed function.
- **#define GPT_TIMEREMAINING_ID ((uint8)0x04U)**
API service ID for Gpt_GetTimeRemaining function.
- **#define GPT_STARTTIMER_ID ((uint8)0x05U)**
API service ID for Gpt_StartTimer function.
- **#define GPT_STOPTIMER_ID ((uint8)0x06U)**
API service ID for Gpt_StopTimer function.
- **#define GPT_ENABLENOTIFICATION_ID ((uint8)0x07U)**
API service ID for Gpt_EnableNotification function.
- **#define GPT_DISABLENOTIFICATION_ID ((uint8)0x08U)**
API service ID for Gpt_DisableNotification function.
- **#define GPT_SETMODE_ID ((uint8)0x09U)**
API service ID for Gpt_SetMode function.
- **#define GPT_DISABLEWAKEUP_ID ((uint8)0x0AU)**
API service ID for Gpt_DisableWakeup function.

- **#define GPT_ENABLEWAKEUP_ID** ((uint8)0x0BU)
API service ID for Gpt_EnableWakeup function.
- **#define GPT_CHECKWAKEUP_ID** ((uint8)0x0CU)
API service ID for Gpt_CheckWakeup function.
- **#define GPT_PROCESSCOMMONINTERRUPT_ID** ((uint8)0x11U)
API service ID for Gpt_ProcessCommonInterrupt generic ISR handler.
- **#define GPT_STARTGLOBALTIME_ID** ((uint8)0x12U)
API service ID for Gpt_StartGlobalTime function.
- **#define GPT_STOPGLOBALTIME_ID** ((uint8)0x13U)
API service ID for Gpt_StopGlobalTime function.
- **#define GPT_CHANGE_NEXT_TIMEOUT_VALUE_ID** ((uint8)0x0FU)
API service ID for Gpt_ChangeNextTimeoutValue function.
- **#define GPT_GET_PREDEF_TIMERVALUE_ID** ((uint8)0x0DU)
API service ID for Gpt_GetPredefTimerValue function.
- **#define GPT_INSTANCE_ID** ((uint8)0U)
Instance ID of this GPT driver.

2.3.2 Macros in Gpt_version.h

- **#define GPT_AR_RELEASE_MAJOR_VERSION** 4
- **#define GPT_AR_RELEASE_MINOR_VERSION** 6
- **#define GPT_AR_RELEASE_REVISION_VERSION** 0
- **#define GPT_MODULE_ID** 100
- **#define GPT_SW_MAJOR_VERSION** 0
- **#define GPT_SW_MINOR_VERSION** 6
- **#define GPT_SW_PATCH_VERSION** 0
- **#define GPT_VENDOR_ID** 174

2.3.3 Macros in Gpt_Cfg.h

- **#define GPT_CFG_AR_RELEASE_MAJOR_VERSION** 4

- **#define GPT_CFG_AR_RELEASE_MINOR_VERSION 6**
- **#define GPT_CFG_AR_RELEASE_REVISION_VERSION 0**
- **#define GPT_CFG_SW_MAJOR_VERSION 0**
- **#define GPT_CFG_SW_MINOR_VERSION 6**
- **#define GPT_CFG_SW_PATCH_VERSION 0**
- **#define GPT_CFG_VENDOR_ID 174**
- **#define GPT_DEV_ERROR_DETECT (STD_ON)**
GPT Dev error detect switch.
- **#define GPT_REPORT_WAKEUP_SOURCE (STD_ON)**
Report Wakeup Source switch.
- **#define GPT_VERSION_INFO_API (STD_ON)**
GPT_VERSION_INFO_API switch.
- **#define GPT_DEINIT_API (STD_ON)**
GPT_DEINIT_API switch.
- **#define GPT_TIME_ELAPSED_API (STD_ON)**
GPT_TIME_ELAPSED_API switch.
- **#define GPT_TIME_REMAINING_API (STD_ON)**
GPT_TIME_REMAINING_API switch.
- **#define GPT_ENABLE_DISABLE_NOTIFICATION_API (STD_ON)**
GPT_ENABLE_DISABLE_NOTIFICATION_API switch.
- **#define GPT_WAKEUP_FUNCTION_API (STD_OFF)**
GPT_WAKEUP_FUNCTION_API switch.
- **#define GPT_PREDEFTIMER_FUNCTION_API (STD_ON)**
GPT_PREDEFTIMER_FUNCTION_API switch.
- **#define GPT_CHANGE_NEXT_TIMEOUT_VALUE (STD_ON)**
Enable/disable support for changing timeout value during timer running.

- **#define GPT_MULTICORE_SUPPORT** (STD_ON)
Enable/disable support for multicore feature.
- **#define GPT_FTU_GTB_SUPPORT** (STD_OFF)
Enable/disable support for Gpt Ftu Gtb feature..
- **#define GPT_PARTITIONS_NB** (3)
The partition number.
- **#define GPT_STANDBY_WAKEUP_SUPPORT** (STD_OFF)
Support for GPT Standby wakeup. The driver shall NOT CLEAR the interrupt flag, the interrupt enable bit and also should not disable the counter, during init *(Gpt_SRtc_Init()) the flag is already set.
- **#define GPT_USER_MODE_SOFTWARE_LOCKED** (STD_OFF)
Enable or disable the access to a hardware register from user mode USER_MODE_SOFTWARE_LOCKED: All reads to HW registers will be done via REG_PROT, user mode access SUPERVISOR_MODE_SOFTWARE_LOCKED: Locks the access to the registers only for supervisor mode.
- **#define GPT_FCPIT_ENABLE_EXT_TRIGGERS** STD_OFF
Enable or disable support external trigger sources.
- **#define GPT_ENABLE_USER_MODE_SUPPORT** (STD_OFF)
- **#define GPT_FTU_MODULE** 0U
- **#define GPT_FCPIT_MODULE** ((GPT_FTU_MODULE_CHANNEL_NUM_U8) * (GPT_FTU_MODULES_NUM_U8))
- **#define GPT_TSTMP_MODULE** (GPT_FCPIT_MODULE + GPT_FCPIT_CHANNEL_NUM)
- **#define GPT_AONTIMER_MODULE** (GPT_TSTMP_MODULE + (GPT_TSTMP_CHANNEL_NUM * GPT_TSTMP_MODULES_NUM))
- **#define GPT_RTC_MODULE** (GPT_AONTIMER_MODULE + GPT_AONTIMER_CHANNEL_NUM)
- **#define GPT_HW_PREDEFTIMER_NUM** (4U)
Number for channels predef timer supported by the platform.
- **#define GPT_HW_CHANNEL_UNUSED** (0xFF)
This define is a reserved logical channel name representing a not existing/configured hardware channel.

- #define **GPT_PRECOMPILE_SUPPORT** (STD_ON)
- #define **GPT_FCPIT_MODULES_NUM** (1U)
- #define **GPT_FCPIT_CHANNEL_NUM** (4U)
- #define **GPT_AONTIMER_MODULES_NUM** (1U)
- #define **GPT_AONTIMER_CHANNEL_NUM** (1U)
- #define **GPT_RTC_MODULES_NUM** (1U)
- #define **GPT_RTC_CHANNEL_NUM** (1U)
- #define **GPT_FTU_MODULES_NUM_U8** (12U)
- #define **GPT_FTU_MODULE_CHANNEL_NUM_U8** (8U)
- #define **MAX_GPT_CHANNEL_NUM** (102U)
The total number of indexes used by the hardware to logic channel map.

2.4 Enums

2.4.1 Enumerations in Gpt.h

2.4.1.1 Gpt_ChannelStatusType

Enumeration	Gpt_ChannelStatusType	
Description	This enumerated type allows the selection of channel status type.	
Values	Value	Description
	GPT_STATUS_UNINITIALIZED = 0	GPT channel status - uninitialized
	GPT_STATUS_INITIALIZED = 1	GPT channel status - initialized
	GPT_STATUS_STOPPED = 2	GPT channel status - stopped
	GPT_STATUS_EXPIRED = 3	GPT channel status - expired
	GPT_STATUS_RUNNING = 4	GPT channel status - running

2.4.2 Enumerations in Gpt_Hw_Types.h

2.4.2.1 Gpt_ModeType

Enumeration	Gpt_ModeType	
Description	This enumerated type allows the selection of different power modes.	
Values	Value	Description
	GPT_MODE_NORMAL = 0U	GPT Normal operation mode of the GPT
	GPT_MODE_SLEEP = 1U	GPT sleep mode

2.4.2.2 Gpt_ChannelModeType

Enumeration	Gpt_ChannelModeType	
Description	GPT channel mode type. Indicates whether the channel mode is "CONTINUOUS" or "ONE SHOT".	
Values	Value	Description
	GPT_CH_MODE_CONTINUOUS = 0U	GPT channel mode - continuous mode
	GPT_CH_MODE_ONESHOT = 1U	GPT channel mode - one-shot mode.

2.4.2.3 Gpt_PredefTimerType

Enumeration	Gpt_PredefTimerType	
Description	N/A	
Values	Value	Description
	GPT_PREDEF_TIMER_1US_16BIT = 0x0U	1us 16bit predef timer
	GPT_PREDEF_TIMER_1US_24BIT	1us 24bit predef timer
	GPT_PREDEF_TIMER_1US_32BIT	1us 32bit predef timer
	GPT_PREDEF_TIMER_100US_32BIT	100us 32bit predef timer

2.4.2.4 Gpt_ModuleType

Enumeration	Gpt_ModuleType	
Description	N/A	
Values	Value	Description
	GPT_MODULE_FTU_0 = 0	FTU module 0
	GPT_MODULE_FTU_1	FTU module 1
	GPT_MODULE_FTU_2	FTU module 2
	GPT_MODULE_FTU_3	FTU module 3
	GPT_MODULE_FTU_4	FTU module 4
	GPT_MODULE_FTU_5	FTU module 5
	GPT_MODULE_FTU_6	FTU module 6
	GPT_MODULE_FTU_7	FTU module 7
	GPT_MODULE_FTU_8	FTU module 8
	GPT_MODULE_FTU_9	FTU module 9
	GPT_MODULE_FTU_10	FTU module 10
	GPT_MODULE_FTU_11	FTU module 11
	GPT_MODULE_FCPIT	FCPIT module
	GPT_MODULE_AONTIMER	AONTIMER module
	GPT_MODULE_RTC	RTC module

2.5 Typedefs

2.5.1 Typedefs in Gpt_Hw_Types.h

- typedef uint8 **Gpt_ChannelType**
GPT channel ID data type.
- typedef uint32 **Gpt_ValueType**
GPT timeout value type. Used for reading and setting the timer value in number of ticks.
- typedef void(* **Gpt_NotificationType**) (void)
GPT channel notification type. The callback notifications shall be configurable as pointers to user defined functions within the configuration structure.
- typedef uint8 **Gpt_PrescalerType**
GPT prescaler data type.
- typedef uint8 **Gpt_ClockSourceType**
GPT clock source data type.

2.6 Structures

2.6.1 Gpt_ConfigType

Structure	Gpt_ConfigType
Description	GPT configuration type. This is the type of the data structure including the configuration set required for initializing the GPT driver.
Diagram	<pre> graph TD Gpt_ConfigType[Gpt_ConfigType] Gpt_ConfigChannelType[Gpt_ConfigChannelType] Gpt_ConfigPredefChannelType[Gpt_ConfigPredefChannelType] Gpt_ConfigChannelCommonType[Gpt_ConfigChannelCommonType] Gpt_ConfigChannelHwSpecificType[Gpt_ConfigChannelHwSpecificType] Gpt_ConfigChannelFtuType[Gpt_ConfigChannelFtuType] Gpt_ConfigChannelAontimerType[Gpt_ConfigChannelAontimerType] Gpt_ConfigChannelFcpitType[Gpt_ConfigChannelFcpitType] Gpt_ConfigType -- pGptChannelConfig --> Gpt_ConfigChannelType Gpt_ConfigType -- pGptChannelPredefConfig --> Gpt_ConfigPredefChannelType Gpt_ConfigChannelType -- tGptChannelCommonConfig --> Gpt_ConfigChannelCommonType Gpt_ConfigChannelType -- tGptChannelHwSpecificConfig --> Gpt_ConfigChannelHwSpecificType Gpt_ConfigChannelCommonType -- tGptFtuChannelConfig --> Gpt_ConfigChannelFtuType Gpt_ConfigChannelCommonType -- tGptAontimerChannelConfig --> Gpt_ConfigChannelAontimerType Gpt_ConfigChannelHwSpecificType -- tGptFcpitChannelConfig --> Gpt_ConfigChannelFcpitType </pre> The diagram illustrates the hierarchical structure of the GPT configuration. At the base is Gpt_ConfigType. It contains pointers to Gpt_ConfigChannelType (via pGptChannelConfig) and Gpt_ConfigPredefChannelType (via pGptChannelPredefConfig). Gpt_ConfigChannelType further contains pointers to Gpt_ConfigChannelCommonType (via tGptChannelCommonConfig) and Gpt_ConfigChannelHwSpecificType (via tGptChannelHwSpecificConfig). Gpt_ConfigChannelCommonType contains pointers to Gpt_ConfigChannelFtuType (via tGptFtuChannelConfig) and Gpt_ConfigChannelAontimerType (via tGptAontimerChannelConfig). Gpt_ConfigChannelHwSpecificType contains a pointer to Gpt_ConfigChannelFcpitType (via tGptFcpitChannelConfig). A legend is provided at the bottom of the diagram.
Data Fields	• const Gpt_ConfigChannelType *pGptChannelConfig Pointer to the GPT channel configuration. • const Gpt_ConfigPredefChannelType *const (*pGptChannelPredefConfig) Pointer to the GPT channel predef timer configuration. • const Gpt_ChannelType *pGptHw2ChannelMap Hardware to logic channel map table. • const uint8 *pGptChannelPartitionMappingPtr

Multicore channel partition map table

2.6.2 Gpt_ChannelInfoType

Structure	Gpt_ChannelInfoType
Description	GPT runtime status type. This is the type of the data structure including the runtime status information of a channel.
Diagram	N/A
Data Fields	• Gpt_ChannelStatusType eChannelStatus GPT channel runtime status. • boolean bNotificationEnabled GPT channel has notification enabled at runtime • boolean bWakeupEnabled GPT channel has wakeup enabled at runtime. • boolean bWakeupGenerated GPT channel executed wakeup.

2.6.3 Gpt_HwChannelInfoType

Structure	Gpt_HwChannelInfoType
Description	GPT channel hardware runtime information. This is the type of the data structure used by the AUTOSAR layer to receive runtime information specific to hardware IP.
Diagram	N/A
Data Fields	• boolean bChannelRollover GPT channel rollover information. • Gpt_ValueType tTargetTime GPT channel target value.

2.6.4 Gpt_ConfigPredefChannelType

Structure	Gpt_ConfigPredefChannelType
Description	Low level configuration for channel specific parameters
Diagram	N/A
Data Fields	• uint8 u8GptPredefChannel GPT HW channel ID. • Gpt_ModuleType eGptPredefModule GPT HW module used. • boolean bGptPredefDebugModeEnable Channel debug mode enable • Gpt_PrescalerType u8GptPredefPrescaler Channel prescaler value. • Gpt_ClockSourceType u8GptPredefClockSource Channel clock source. • uint32 u32MaxValue Counter max value.

2.6.5 Gpt_ConfigChannelCommonType

Structure	Gpt_ConfigChannelCommonType
Description	GPT common configuration.
Diagram	N/A
Data Fields	• uint8 u8GptChannel GPT HW channel ID. • Gpt_ModuleType eGptModule GPT HW module used. • Gpt_ChannelModeType eGptChannelMode GPT channel mode. • Gpt_ValueType u32GptChannelTickValueMax Channel max tick value. • boolean bGptDebugModeEnable Channel debug mode enable. • uint32 u32GptWakeupSource EcuM wake up source ID. • boolean bGptEnableWakeup GPT ch WakeUp enable. • Gpt_NotificationType pGptNotification Pointer to external callback.

2.6.6 Gpt_ConfigChannelHwSpecificType

Structure	Gpt_ConfigChannelHwSpecificType
Description	GPT hardware specific configuration.
Diagram	N/A
Data Fields	• Gpt_ConfigChannelFtuType tGptFtuChannelConfig FTU channel configuration. • Gpt_ConfigChannelAontimerType tGptAontimerChannelConfig AONTIMER channel configuration. • Gpt_ConfigChannelFcptType tGptFcptChannelConfig FCPIT channel configuration.

2.6.7 Gpt_ConfigChannelType

Structure	Gpt_ConfigChannelType
Description	GPT channel configuration
Diagram	N/A
Data Fields	• Gpt_ConfigChannelCommonType tGptChannelCommonConfig GPT channel common configuration. • Gpt_ConfigChannelHwSpecificType tGptChannelHwSpecificConfig GPT channel specific configuration.

2.7 API Functions

2.7.1 Functions in Gpt.h

2.7.1.1 void Gpt_ChangeNextTimeoutValue (Gpt_ChannelType *Channel*, Gpt_ValueType *Value*)

Function	void Gpt_ChangeNextTimeoutValue (Gpt_ChannelType <i>Channel</i> , Gpt_ValueType <i>Value</i>)	
Description	Change the next timeout value.	
Diagram	<pre> graph LR Gpt_ChangeNextTimeoutValue[Gpt_ChangeNextTimeoutValue] --> Gpt_HL_ChangeNextTimeoutValue[Gpt_HL_ChangeNextTimeoutValue] Gpt_HL_ChangeNextTimeoutValue --> Gpt_LL_Ftu_ChangeNextTimeoutValue[Gpt_LL_Ftu_ChangeNextTimeoutValue] Gpt_ChangeNextTimeoutValue --> Gpt_ReportDetError[Gpt_ReportDetError] Gpt_ChangeNextTimeoutValue --> Gpt_ReportRuntimeDetError[Gpt_ReportRuntimeDetError] </pre>	
Parameters	Parameter	Description
	Channel	Logic channel number
	Value	Channel value
Returns	N/A	

2.7.1.2 void Gpt_CheckWakeup (EcuM_WakeupSourceType *WakeupSource*)

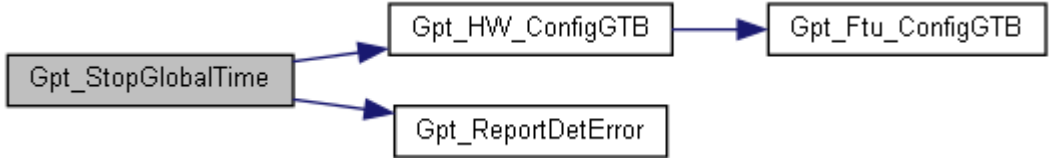
Function	void Gpt_CheckWakeup (EcuM_WakeupSourceType <i>WakeupSource</i>)	
Description	Check the wakeup status.	
Diagram	<pre> graph LR Gpt_CheckWakeup[Gpt_CheckWakeup] --> Gpt_ReportDetError[Gpt_ReportDetError] </pre>	
Parameters	Parameter	Description
	WakeupSource	GPT wakeup source type
Returns	N/A	

2.7.1.3 void Gpt_StartGlobalTime(const uint32 u32Group)

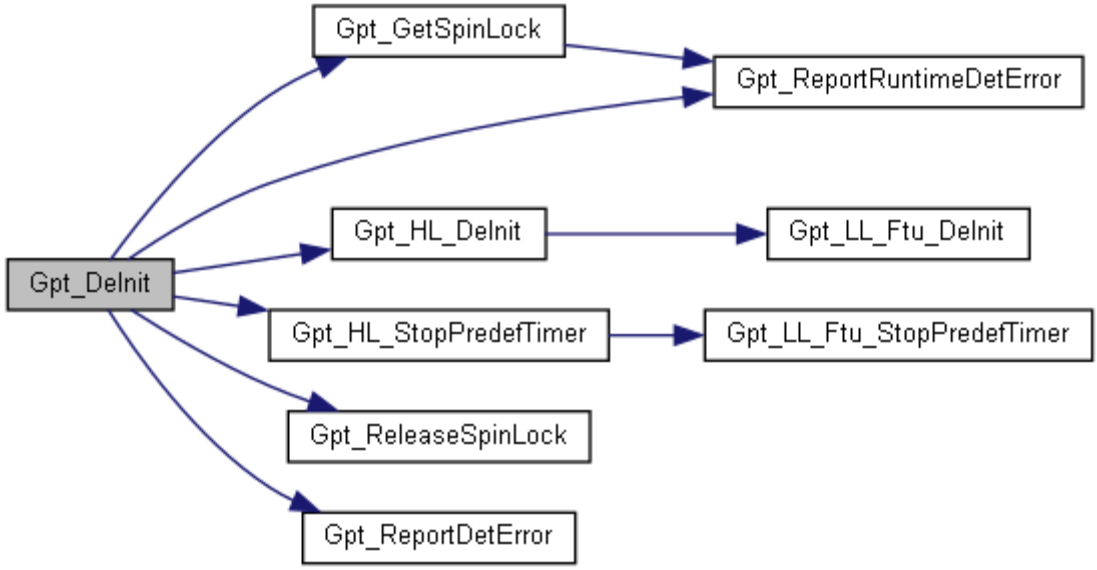
Function	void Gpt_StartGlobalTime(const uint32 u32Group)	
Description	Enable the Ftu Gtb function	
Diagram	<pre> graph LR Gpt_StartGlobalTime[Gpt_StartGlobalTime] --> Gpt_HWV_ConfigGTB[Gpt_HWV_ConfigGTB] Gpt_HWV_ConfigGTB --> Gpt_Ftu_ConfigGTB[Gpt_Ftu_ConfigGTB] Gpt_StartGlobalTime --> Gpt_ReportDetError[Gpt_ReportDetError] </pre>	
Parameters	Parameter	Description
	u32Group	Ftu Gtb Group
Returns	N/A	

2.7.1.4 void Gpt_StopGlobalTime(const uint32 u32Group)

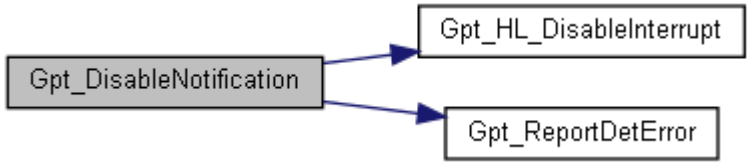
Function	void Gpt_StopGlobalTime(const uint32 u32Group)	
Description	Disable the Ftu Gtb function	

Diagram	 <pre> graph LR Gpt_StopGlobalTime --> Gpt_HW_ConfigGTB Gpt_StopGlobalTime --> Gpt_ReportDetError Gpt_HW_ConfigGTB --> Gpt_Ftu_ConfigGTB </pre>	
Parameters	Parameter	Description
	u32Group	Ftu Gtb Group
Returns	N/A	

2.7.1.5 void Gpt_Delnit (void)

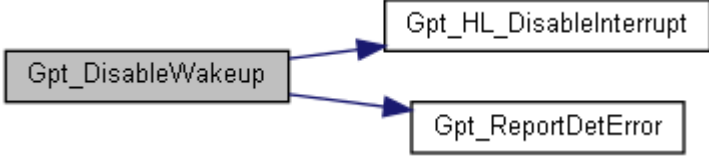
Function	void Gpt_Delnit (void)	
Description	De-initialization.	
Diagram	 <pre> graph LR Gpt_Delnit --> Gpt_GetSpinLock Gpt_Delnit --> Gpt_ReportRuntimeDetError Gpt_Delnit --> Gpt_HL_Delnit Gpt_Delnit --> Gpt_LL_Ftu_Delnit Gpt_Delnit --> Gpt_HL_StopPredefTimer Gpt_Delnit --> Gpt_LL_Ftu_StopPredefTimer Gpt_Delnit --> Gpt_ReleaseSpinLock Gpt_Delnit --> Gpt_ReportDetError Gpt_HL_Delnit --> Gpt_LL_Ftu_Delnit Gpt_HL_StopPredefTimer --> Gpt_LL_Ftu_StopPredefTimer </pre>	
Parameters	N/A	
Returns	N/A	

2.7.1.6 void Gpt_DisableNotification (Gpt_ChannelType Channel)

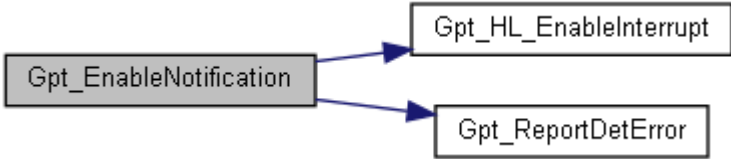
Function	void Gpt_DisableNotification (Gpt_ChannelType Channel)	
Description	Disable channel notification.	
Diagram	 <pre> graph LR Gpt_DisableNotification --> Gpt_HL_DisableInterrupt Gpt_DisableNotification --> Gpt_ReportDetError </pre>	
Parameters	Parameter	Description
	Channel	Logic channel number
Returns	N/A	

2.7.1.7 void Gpt_DisableWakeup (Gpt_ChannelType Channel)

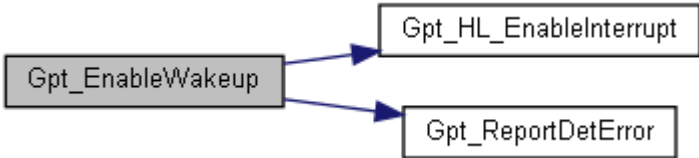
Function	void Gpt_DisableWakeup (Gpt_ChannelType Channel)	
Description	Disable channel wakeup.	

Diagram		
Parameters	Parameter	Description
	Channel	Logic channel number
Returns	N/A	

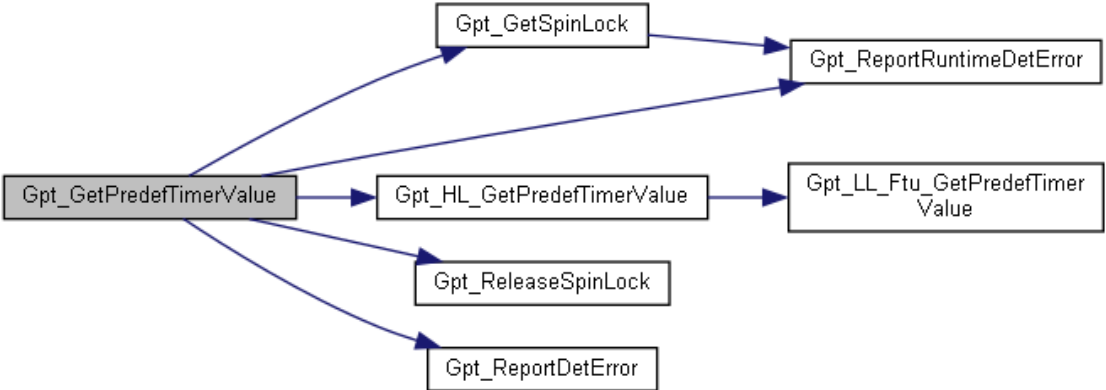
2.7.1.8 void Gpt_EnableNotification (Gpt_ChannelType Channel)

Function	void Gpt_EnableNotification (Gpt_ChannelType Channel)	
Description	Enable channel notification.	
Diagram		
Parameters	Parameter	Description
	Channel	Logic channel number
Returns	N/A	

2.7.1.9 void Gpt_EnableWakeup (Gpt_ChannelType Channel)

Function	void Gpt_EnableWakeup (Gpt_ChannelType Channel)	
Description	Enable the channel wakeup.	
Diagram		
Parameters	Parameter	Description
	Channel	Logic channel number
Returns	N/A	

2.7.1.10 Std_ReturnType Gpt_GetPredefTimerValue (Gpt_PredefTimerType PredefTimer, uint32 * TimeValuePtr)

Function	Std_ReturnType Gpt_GetPredefTimerValue (Gpt_PredefTimerType PredefTimer, uint32 * TimeValuePtr)	
Description	Get the Predef timer value.	
Diagram		

Parameters	Parameter	Description
	PredefTimer	Predef timer enumeration
	TimeValuePtr	Time value pointer
Returns	Std_ReturnType API status return type.	

2.7.1.11 Gpt_ValueType Gpt_GetTimeElapsed (Gpt_ChannelType Channel)

Function	Gpt_ValueType Gpt_GetTimeElapsed (Gpt_ChannelType Channel)	
Description	Get the channel elapsed value.	
Diagram	<pre> graph LR Gpt_GetTimeElapsed --> Gpt_HL_GetTimeElapsed Gpt_HL_GetTimeElapsed --> Gpt_LL_Ftu_GetTimeElapsed Gpt_GetTimeElapsed --> Gpt_ReportDetError </pre>	
Parameters	Parameter	Description
	Channel	Logic channel number
Returns	Gpt_ValueType	

2.7.1.12 Gpt_ValueType Gpt_GetTimeRemaining (Gpt_ChannelType Channel)

Function	Gpt_ValueType Gpt_GetTimeRemaining (Gpt_ChannelType Channel)	
Description	Get the channel remaining value.	
Diagram	<pre> graph LR Gpt_GetTimeRemaining --> Gpt_HL_GetTimeElapsed Gpt_HL_GetTimeElapsed --> Gpt_LL_Ftu_GetTimeElapsed Gpt_GetTimeRemaining --> Gpt_ReportDetError </pre>	
Parameters	Parameter	Description
	Channel	Logic channel number
Returns	Gpt_ValueType	

2.7.1.13 void Gpt_GetVersionInfo (Std_VersionInfoType * VersionInfoPtr)

Function	void Gpt_GetVersionInfo (Std_VersionInfoType * VersionInfoPtr)	
Description	Get the API of the GPT module version information.	
Diagram	N/A	
Parameters	Parameter	Description
	VersionInfoPtr	Version information structure pointer
Returns	N/A	

2.7.1.14 void Gpt_Init (const Gpt_ConfigType * ConfigPtr)

Function	void Gpt_Init (const Gpt_ConfigType * ConfigPtr)	
Description	GPT initialization.	

Diagram	<pre> graph LR Gpt_Init --> Gpt_GetSpinLock Gpt_Init --> Gpt_HL_Init Gpt_Init --> Gpt_HL_StartPredefTimer Gpt_Init --> Gpt_ReleaseSpinLock Gpt_Init --> Gpt_ReportDetError Gpt_GetSpinLock --> Gpt_ReportRuntimeDetError Gpt_HL_Init --> Gpt_LL_Ftu_Init Gpt_HL_StartPredefTimer --> Gpt_LL_Ftu_StartPredefTimer </pre>	
Parameters	Parameter	Description
	ConfigPtr	Pointer to configuration value
Returns	N/A	

2.7.1.15 void Gpt_SetMode (Gpt_ModeType Mode)

Function	void Gpt_SetMode (Gpt_ModeType Mode)	
Description	Set the GPT mode.	
Diagram	<pre> graph LR Gpt_SetMode --> Gpt_GetSpinLock Gpt_SetMode --> Gpt_HL_DisableInterrupt Gpt_SetMode --> Gpt_HL_EnableInterrupt Gpt_SetMode --> Gpt_HL_GetTimeElapsed Gpt_SetMode --> Gpt_HL_StartPredefTimer Gpt_SetMode --> Gpt_HL_StopPredefTimer Gpt_SetMode --> Gpt_HL_StopTimer Gpt_SetMode --> Gpt_ReleaseSpinLock Gpt_SetMode --> Gpt_ValidSetMode Gpt_GetSpinLock --> Gpt_ReportRuntimeDetError Gpt_HL_GetTimeElapsed --> Gpt_LL_Ftu_GetTimeElapsed Gpt_HL_StartPredefTimer --> Gpt_LL_Ftu_StartPredefTimer Gpt_HL_StopPredefTimer --> Gpt_LL_Ftu_StopPredefTimer Gpt_HL_StopTimer --> Gpt_LL_Ftu_StopTimer Gpt_ValidSetMode --> Gpt_ReportDetError </pre>	
Parameters	Parameter	Description
	Mode	GPT module mode.
Returns	N/A	

2.7.1.16 void Gpt_StartTimer (Gpt_ChannelType Channel, Gpt_ValueType Value)

Function	void Gpt_StartTimer (Gpt_ChannelType Channel, Gpt_ValueType Value)	
Description	Start a channel.	

Diagram	<pre>graph LR; Gpt_StartTimer[Gpt_StartTimer] --> Gpt_HL_EnableInterrupt[Gpt_HL_EnableInterrupt]; Gpt_StartTimer --> Gpt_HL_StartTimer[Gpt_HL_StartTimer]; Gpt_HL_StartTimer --> Gpt_LL_Ftu_StartTimer[Gpt_LL_Ftu_StartTimer]; Gpt_StartTimer --> Gpt_ReportDetError[Gpt_ReportDetError]; Gpt_StartTimer --> Gpt_ReportRuntimeDetError[Gpt_ReportRuntimeDetError];</pre>	
Parameters	Parameter	Description
	Channel	Logic channel number
	Value	Start time value
Returns	N/A	

2.7.1.17 void Gpt_StopTimer (Gpt_ChannelType Channel)

Function	void Gpt_StopTimer (Gpt_ChannelType <i>Channel</i>)					
Description	Stop channel.					
Diagram	<pre>graph LR; Gpt_StopTimer[Gpt_StopTimer] --> Gpt_HL_GetTimeElapsed[Gpt_HL_GetTimeElapsed]; Gpt_HL_GetTimeElapsed --> Gpt_LL_Ftu_GetTimeElapsed[Gpt_LL_Ftu_GetTimeElapsed]; Gpt_StopTimer --> Gpt_HL_StopTimer[Gpt_HL_StopTimer]; Gpt_HL_StopTimer --> Gpt_LL_Ftu_StopTimer[Gpt_LL_Ftu_StopTimer]; Gpt_StopTimer --> Gpt_ReportDetError[Gpt_ReportDetError];</pre>					
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>Channel</td><td>Logic channel number</td></tr></table>	Parameter	Description	Channel	Logic channel number	
Parameter	Description					
Channel	Logic channel number					
Returns	N/A					

2.7.1.18 void Gpt_ProcessCommonInterrupt (uint8 u8MapTableIndex)

Function	void Gpt_ProcessCommonInterrupt (uint8 u8MapTableIndex)	
Description	Handle common function in interrupt service routine, such as switch channel status.	
Diagram	N/A	
Parameters	Parameter	Description
	u8MapTableIndex	Index of module and channel map table
Returns	N/A	

2.8 Hardware Functions

2.8.1 Functions in Gpt_Hw.h

2.8.1.1 Std_ReturnType Gpt_HL_ChangeNextTimeoutValue (const Gpt_ConfigChannelType * pChannelConfig, Gpt_ValueType u32Value)

Function	Std_ReturnType Gpt_HL_ChangeNextTimeoutValue (const Gpt_ConfigChannelType * pChannelConfig, Gpt_ValueType u32Value)
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Description	Change the next timeout value.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer
	u32Value	Next timeout value
Returns	Std_ReturnType Function return type.	
Referenced By	Gpt_ChangeNextTimeoutValue().	

2.8.1.2 void Gpt_HL_DeInit (const Gpt_ConfigChannelType * pChannelConfig)

Function	void Gpt_HL_DeInit (const Gpt_ConfigChannelType * pChannelConfig)	
Description	De-initialization.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer
Returns	N/A	
Referenced By	Gpt_DeInit()	

2.8.1.3 void Gpt_HL_DisableInterrupt (const Gpt_ConfigChannelType * pChannelConfig)

Function	void Gpt_HL_DisableInterrupt (const Gpt_ConfigChannelType * pChannelConfig)	
Description	Disable an interrupt.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer.
Returns	N/A	
Referenced By	Gpt_DisableNotification(), Gpt_DisableWakeup(), and Gpt_SetMode().	

2.8.1.4 void Gpt_HL_EnableInterrupt (const Gpt_ConfigChannelType * pChannelConfig)

Function	void Gpt_HL_EnableInterrupt (const Gpt_ConfigChannelType * pChannelConfig)	
Description	Enable an interrupt.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer.
Returns	N/A	
Referenced By	Gpt_EnableNotification(), Gpt_EnableWakeup(), Gpt_SetMode(), and Gpt_StartTimer().	

2.8.1.5 void Gpt_HL_GetPredefTimerValue (const Gpt_ConfigPredefChannelType * pChannelConfig, uint32 * u32TimeValuePtr)

Function	void Gpt_HL_GetPredefTimerValue (const Gpt_ConfigPredefChannelType * pChannelConfig, uint32 * u32TimeValuePtr)	
Description	Get the predef timer value.	
Parameters	Parameter	Description
	pChannelConfig	Predef timer configuration pointer.
	u32TimeValuePtr	Timer value pointer.
Returns	N/A	
Referenced By	Gpt_GetPredefTimerValue()	

2.8.1.6 Gpt_ValueType Gpt_HL_GetTimeElapsed (const Gpt_ConfigChannelType * pChannelConfig, Gpt_HwChannelInfoType * pChannelInfo)

Function	Gpt_ValueType Gpt_HL_GetTimeElapsed (const Gpt_ConfigChannelType * pChannelConfig, Gpt_HwChannelInfoType * pChannelInfo)	
Description	Get the timer elapsed.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer.
	pChannelInfo	Channel information pointer.
Returns	Gpt_ValueType Elapsed value	
Referenced By	Gpt_GetTimeElapsed(), Gpt_GetTimeRemaining(), Gpt_SetMode(), and Gpt_StopTimer()	

2.8.1.7 void Gpt_HL_Init (const Gpt_ConfigChannelType * pChannelConfig)

Function	void Gpt_HL_Init (const Gpt_ConfigChannelType * pChannelConfig)	
Description	Initialize a channel.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer
Returns	N/A	
Referenced By	Gpt_Init()	

2.8.1.8 void Gpt_HL_StartPredefTimer (const Gpt_ConfigType * pConfigPtr)

Function	void Gpt_HL_StartPredefTimer (const Gpt_ConfigType * pConfigPtr)	
Description	Start the predef timer.	
Parameters	Parameter	Description
	pConfigPtr	Predef configuration pointer.
Returns	N/A	
Referenced By	Gpt_Init(), and Gpt_SetMode()	

2.8.1.9 void Gpt_HW_ConfigGTB(const uint32 u32Group,boolean benable)

Function	void Gpt_HW_ConfigGTB(const uint32 u32Group,boolean benable)	
Description	Config the FTU GTB Function	
Parameters	Parameter	Description
	u32Group	FTU GTB Group
	benable	enable/disable FtU Gtb function.
Returns	N/A	
Referenced By	Gpt_StartGlobalTime (), and Gpt_StopGlobalTime ()	

2.8.1.10 void Gpt_HL_StartTimer (const Gpt_ConfigChannelType * pChannelConfig, Gpt_ValueType u32Value)

Function	void Gpt_HL_StartTimer (const Gpt_ConfigChannelType * pChannelConfig, Gpt_ValueType u32Value)	
Description	Start the timer.	
Parameters	Parameter	Description

	pChannelConfig	Channel configuration pointer.
	u32Value	Start timer value.
Returns	N/A	
Referenced By	Gpt_StartTimer()	

2.8.1.11 void Gpt_HL_StopPredefTimer (const Gpt_ConfigType * pConfigPtr)

Function	void Gpt_HL_StopPredefTimer (const Gpt_ConfigType * pConfigPtr)	
Description	Stop the predef timer.	
Parameters	Parameter	Description
	pConfigPtr	GPT configuration pointer.
Returns	N/A	
Referenced By	Gpt_DelInit(), and Gpt_SetMode()	

2.8.1.12 void Gpt_HL_StopTimer (const Gpt_ConfigChannelType * pChannelConfig)

Function	void Gpt_HL_StopTimer (const Gpt_ConfigChannelType * pChannelConfig)	
Description	Stop the timer.	
Parameters	Parameter	Description
	pChannelConfig	Channel configuration pointer.
Returns	N/A	
Referenced By	Gpt_ProcessCommonInterrupt(), Gpt_SetMode(), and Gpt_StopTimer()	

2.8.2 Functions in Gpt_Aontimer.h

2.8.2.1 void Gpt_LL_Aontimer_DelInit (void)

Function	void Gpt_LL_Aontimer_DelInit (void)
Description	AONTIMER start timer.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_HL_DelInit()

2.8.2.2 void Gpt_LL_Aontimer_DisableInterrupt (void)

Function	void Gpt_LL_Aontimer_DisableInterrupt (void)
Description	AONTIMER disable interrupt.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_HL_DisableInterrupt()

2.8.2.3 void Gpt_LL_Aontimer_EnableInterrupt (void)

Function	void Gpt_LL_Aontimer_EnableInterrupt (void)
Description	AONTIMER enable interrupt.
Parameters	N/A

Returns	N/A
Referenced By	Gpt_HL_EnableInterrupt()

2.8.2.4 uint32 Gpt_LL_Aontimer_GetPredefTimerValue (void)

Function	uint32 Gpt_LL_Aontimer_GetPredefTimerValue (void)
Description	AONTIMER get predef timer value.
Parameters	N/A
Returns	uint32 Predef timer value
Referenced By	Gpt_HL_GetPredefTimerValue()

2.8.2.5 uint32 Gpt_LL_Aontimer_GetTimeElapsed (boolean * pbReturnChannelRollover, uint32 * pu32TargetValue)

Function	uint32 Gpt_LL_Aontimer_GetTimeElapsed (boolean * pbReturnChannelRollover, uint32 * pu32TargetValue)	
Description	AONTIMER get elapsed value.	
Parameters	Parameter	Description
	pbReturnChannelRollover	Channel roll-over flag
	pu32TargetValue	AONTIMER target value
Returns	uint32 Elapsed value	
Referenced By	Gpt_HL_GetTimeElapsed()	

2.8.2.6 void Gpt_LL_Aontimer_Init (const Gpt_ConfigChannelAontimerType *const pAontiemrConf)

Function	void Gpt_LL_Aontimer_Init (const Gpt_ConfigChannelAontimerType *const pAontiemrConf)	
Description	AONTIMER initialization.	
Parameters	Parameter	Description
	pAontiemrConf	AONTIMER initialization configuration pointer.
Returns	N/A	
Referenced By	Gpt_HL_Init()	

2.8.2.7 void Gpt_LL_Aontimer_StartPredefTimer (const Gpt_AontimerClkSrcType eClkSrc, const Gpt_AontimerPrescalerType ePrescaler)

Function	void Gpt_LL_Aontimer_StartPredefTimer (const Gpt_AontimerClkSrcType eClkSrc, const Gpt_AontimerPrescalerType ePrescaler)	
Description	AONTIMER start predef timer.	
Parameters	Parameter	Description
	eClkSrc	AONTIMER clock source.
	ePrescaler	AONTIMER timer prescaler.
Returns	N/A	
Referenced By	Gpt_HL_StartPredefTimer()	

2.8.2.8 void Gpt_LL_Aontimer_StartTimer (uint16 u16Value)

Function	void Gpt_LL_Aontimer_StartTimer (uint16 u16Value)
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Description	AONTIMER start timer.	
Parameters	Parameter	Description
	u16Value	Start timer value
Returns	N/A	
Referenced By	Gpt_HL_StartTimer()	

2.8.2.9 void Gpt_LL_Aontimer_StopPredefTimer (void)

Function	void Gpt_LL_Aontimer_StopPredefTimer (void)	
Description	Stop the predef timer.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_StopPredefTimer()	

2.8.2.10 void Gpt_LL_Aontimer_StopTimer (void)

Function	void Gpt_LL_Aontimer_StopTimer (void)	
Description	AONTIMER stop timer.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_StopTimer()	

2.8.3 Functions in Gpt_FCPit.h

2.8.3.1 void Gpt_LL_FCPit_DeInit (Gpt_FcpitChannelType *eChannel*)

Function	void Gpt_LL_FCPit_DeInit (Gpt_FcpitChannelType <i>eChannel</i>)	
Description	De-initialization.	
Parameters	Parameter	Description
	eChannel	FCPIT channel
Returns	N/A	
Referenced By	Gpt_HL_DeInit()	

2.8.3.2 void Gpt_LL_FCPIT_DisableInterrupt (Gpt_FcpitChannelType *eChannel*)

Function	void Gpt_LL_FCPIT_DisableInterrupt (Gpt_FcpitChannelType <i>eChannel</i>)	
Description	Disable an FCPIT interrupt.	
Parameters	Parameter	Description
	eChannel	FCPIT channel
Returns	N/A	
Referenced By	Gpt_HL_DisableInterrupt()	

2.8.3.3 void Gpt_LL_FCPit_EnableInterrupt (Gpt_FcpitChannelType *eChannel*)

Function	void Gpt_LL_FCPit_EnableInterrupt (Gpt_FcpitChannelType <i>eChannel</i>)	
Description	Enable an FCPIT interrupt.	

Parameters	Parameter	Description
	eChannel	FCPIT channel
Returns	N/A	
Referenced By	Gpt_HL_EnableInterrupt()	

2.8.3.4 uint32 Gpt_LL_FCPIT_GetPredefTimerValue (Gpt_FcpitChannelType eChannel, uint32 u32MaxValue)

Function	uint32 Gpt_LL_FCPIT_GetPredefTimerValue (Gpt_FcpitChannelType eChannel, uint32 u32MaxValue)	
Description	Get the predef timer value.	
Parameters	Parameter	Description
	eChannel	FCPIT channel
	u32MaxValue	FCPIT max value
Returns	FCPIT predef timer value.	
Referenced By	Gpt_HL_GetPredefTimerValue()	

2.8.3.5 uint32 Gpt_LL_FCPit_GetTimeElapsed (Gpt_FcpitChannelType eChannel, boolean * pbReturnChannelRollover, uint32 * pu32TargetValue)

Function	uint32 Gpt_LL_FCPit_GetTimeElapsed (Gpt_FcpitChannelType eChannel, boolean * pbReturnChannelRollover, uint32 * pu32TargetValue)	
Description	Get the FCPIT elapsed value.	
Parameters	Parameter	Description
	eChannel	FCPIT channel
	pbReturnChannelRollover	Channel roll-over flag
	pu32TargetValue	FCPIT target value
Returns	uint32 Elapsed value	
Referenced By	Gpt_HL_GetTimeElapsed()	

2.8.3.6 void Gpt_LL_Fcpit_Init (Gpt_FcpitChannelType eChannel, boolean bDbgMode, const Gpt_ConfigChannelFcpitType *const pFcpitConfig)

Function	void Gpt_LL_Fcpit_Init (Gpt_FcpitChannelType eChannel, boolean bDbgMode, const Gpt_ConfigChannelFcpitType *const pFcpitConfig)	
Description	Initialize the FCPIT timer.	
Parameters	Parameter	Description
	eChannel	FCPIT channel.
	bDbgMode	FCPIT debug mode flag.
	pFcpitConfig	FCPIT initialization configuration pointer.
Returns	N/A	
Referenced By	Gpt_HL_Init()	

2.8.3.7 void Gpt_LL_FCPIT_StartPredefTimer (Gpt_FcpitChannelType eChannel, boolean bDbgModeEnable, uint32 u32MaxValue)

Function	void Gpt_LL_FCPIT_StartPredefTimer (Gpt_FcpitChannelType eChannel, boolean bDbgModeEnable, uint32 u32MaxValue)	
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Description	Start the predef timer.	
Parameters	Parameter	Description
	eChannel	FCPIT channel.
	bDbgModeEnable	Enable/Disable debug mode flag.
	u32MaxValue	FCPIT max value.
Returns	N/A	
Referenced By	Gpt_HL_StartPredefTimer()	

2.8.3.8 void Gpt_LL_FCPit_StartTimer (Gpt_FcpitChannelType eChannel, uint32 u32Value)

Function	void Gpt_LL_FCPit_StartTimer (Gpt_FcpitChannelType eChannel, uint32 u32Value)	
Description	Start the FCPIT timer.	
Parameters	Parameter	Description
	eChannel	FCPIT channel.
	u32Value	FCPIT start value.
Returns	N/A	
Referenced By	Gpt_HL_StartTimer()	

2.8.3.9 void Gpt_LL_FCPIT_StopPredefTimer (Gpt_FcpitChannelType eChannel)

Function	void Gpt_LL_FCPIT_StopPredefTimer (Gpt_FcpitChannelType eChannel)	
Description	Stop the predef timer.	
Parameters	Parameter	Description
	eChannel	FCPIT channel.
Returns	N/A	
Referenced By	Gpt_HL_StopPredefTimer()	

2.8.3.10 void Gpt_LL_FCPit_StopTimer (Gpt_FcpitChannelType eChannel)

Function	void Gpt_LL_FCPit_StopTimer (Gpt_FcpitChannelType eChannel)	
Description	Stop the FCPIT timer.	
Parameters	Parameter	Description
	eChannel	FCPIT channel.
Returns	N/A	
Referenced By	Gpt_HL_StopTimer()	

2.8.4 Functions in Gpt_Ftu.h

2.8.4.1 void Gpt_LL_Ftu_ChangeNextTimeoutValue (Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32Value)

Function	void Gpt_LL_Ftu_ChangeNextTimeoutValue (Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32Value)	
Description	FTU change the next timeout value.	
Parameters	Parameter	Description
	eModule	FTU module.
	u8Channel	FTU channel for current module.

	u32Value	The value of next timeout.
Returns	N/A	
Referenced By	Gpt_HL_ChangeNextTimeoutValue()	

2.8.4.2 void Gpt_LL_Ftu_DeInit (Gpt_ModuleType eModule, uint8 u8Channel)

Function	void Gpt_LL_Ftu_DeInit (Gpt_ModuleType eModule, uint8 u8Channel)	
Description	De-initialization.	
Parameters	Parameter	Description
	eModule	FTU module.
	u8Channel	FTU channel for current module.
Returns	N/A	
Referenced By	Gpt_HL_DeInit()	

2.8.4.3 uint32 Gpt_LL_Ftu_GetPredefTimerValue (Gpt_ModuleType eModule)

Function	uint32 Gpt_LL_Ftu_GetPredefTimerValue (Gpt_ModuleType eModule)	
Description	Get the predef timer value.	
Parameters	Parameter	Description
	eModule	FTU module.
Returns	uint32 Predef timer value	
Referenced By	Gpt_HL_GetPredefTimerValue()	

2.8.4.4 uint32 Gpt_LL_Ftu_GetTimeElapsed (Gpt_ModuleType eModule, uint8 u8Channel, boolean * pReturnChannelOverflow, uint32 * pTargetValue)

Function	uint32 Gpt_LL_Ftu_GetTimeElapsed (Gpt_ModuleType eModule, uint8 u8Channel, boolean * pReturnChannelOverflow, uint32 * pTargetValue)	
Description	Get the FTU elapsed value.	
Parameters	Parameter	Description
	eModule	FTU module.
	u8Channel	FTU channel for the current module.
	pReturnChannelOverflow	Channel overflow flag.
	pTargetValue	Target value
Returns	uint32 Elapsed value	
Referenced By	Gpt_HL_GetTimeElapsed()	

2.8.4.5 void Gpt_Ftu_ConfigGTB(const uint32 u32Group,boolean benable)

Function	void Gpt_Ftu_ConfigGTB(const uint32 u32Group,boolean benable)	
Description	FTU initialization.	
Parameters	Parameter	Description
	u32Group	FTU GTB Group
	benable	enable/disable Ftu Gtb function.
Returns	N/A	
Referenced By	Gpt_HW_ConfigGTB ()	

2.8.4.6 `void Gpt_LL_Ftu_Init (Gpt_ModuleType eModule, uint8 u8Channel, boolean bDbgModeEnable, Gpt_ConfigChannelFtuType * pFtuConfig)`

Function	void Gpt_LL_Ftu_Init (Gpt_ModuleType <i>eModule</i> , uint8 <i>u8Channel</i> , boolean <i>bDbgModeEnable</i> , Gpt_ConfigChannelFtuType * <i>pFtuConfig</i>)	
Description	FTU initialization.	
Parameters	Parameter	Description
	<i>eModule</i>	FTU module.
	<i>u8Channel</i>	FTU channel for the current module.
	<i>bDbgModeEnable</i>	Enable/Disable debug mode flag.
	<i>pFtuConfig</i>	FTU configuration pointer.
Returns	N/A	
Referenced By	Gpt_HL_Init()	

2.8.4.7 `void Gpt_LL_Ftu_StartPredefTimer (Gpt_ModuleType eModule, uint8 u8Channel, boolean bDbgModeEnable, Gpt_FtuModuleClkSrcType eClkSrc, Gpt_FtuPrescalerType ePrescale)`

Function	void Gpt_LL_Ftu_StartPredefTimer (Gpt_ModuleType <i>eModule</i> , uint8 <i>u8Channel</i> , boolean <i>bDbgModeEnable</i> , Gpt_FtuModuleClkSrcType <i>eClkSrc</i> , Gpt_FtuPrescalerType <i>ePrescale</i>)	
Description	Start the predef timer.	
Parameters	Parameter	Description
	<i>eModule</i>	FTU module.
	<i>u8Channel</i>	FTU channel for the current module.
	<i>bDbgModeEnable</i>	Enable/Disable debug mode flag.
	<i>eClkSrc</i>	FTU clock source.
	<i>ePrescale</i>	FTU prescaler.
Returns	N/A	
Referenced By	Gpt_HL_StartPredefTimer()	

2.8.4.8 `void Gpt_LL_Ftu_StartTimer (Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32Value)`

Function	void Gpt_LL_Ftu_StartTimer (Gpt_ModuleType <i>eModule</i> , uint8 <i>u8Channel</i> , uint32 <i>u32Value</i>)	
Description	Start the FTU timer.	
Parameters	Parameter	Description
	<i>eModule</i>	FTU module.
	<i>u8Channel</i>	FTU channel for the current module.
	<i>u32Value</i>	FTU start timer value.
Returns	N/A	
Referenced By	Gpt_HL_StartTimer()	

2.8.4.9 `void Gpt_LL_Ftu_StopPredefTimer (Gpt_ModuleType eModule, uint8 u8Channel)`

Function	void Gpt_LL_Ftu_StopPredefTimer (Gpt_ModuleType <i>eModule</i> , uint8 <i>u8Channel</i>)
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Description	Stop the FTU predef timer.	
Parameters	Parameter	Description
	eModule	FTU module.
	u8Channel	FTU channel for the current module.
Returns	N/A	
Referenced By	Gpt_HL_StopPredefTimer()	

2.8.4.10 void Gpt_LL_Ftu_StopTimer (Gpt_ModuleType eModule, uint8 u8Channel)

Function	void Gpt_LL_Ftu_StopTimer (Gpt_ModuleType eModule, uint8 u8Channel)	
Description	Stop the FTU timer.	
Parameters	Parameter	Description
	eModule	FTU module.
	u8Channel	FTU channel for the current module.
Returns	N/A	
Referenced By	Gpt_HL_StopTimer()	

2.8.5 Functions in Gpt_Rtc.h

2.8.5.1 void Gpt_LL_Rtc_DeInit (void)

Function	void Gpt_LL_Rtc_DeInit (void)	
Description	De-initialization.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_DeInit()	

2.8.5.2 void Gpt_LL_Rtc_DisableInterrupt (void)

Function	void Gpt_LL_Rtc_DisableInterrupt (void)	
Description	RTC disable interrupt.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_DisableInterrupt()	

2.8.5.3 void Gpt_LL_Rtc_EnableInterrupt (void)

Function	void Gpt_LL_Rtc_EnableInterrupt (void)	
Description	RTC enable interrupt.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_EnableInterrupt()	

2.8.5.4 uint32 Gpt_LL_Rtc_GetTimeElapsed (boolean * pReturnChannelRollover, uint32 * pTargetValue)

Function	uint32 Gpt_LL_Rtc_GetTimeElapsed (boolean * pReturnChannelRollover, uint32 * pTargetValue)	
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Description	Get the timer elapsed value.	
Parameters	Parameter	Description
	pReturnChannelRollover	RTC channel roll-over flag
	pTargetValue	Timer target value
Returns	uint32 Elapsed value	
Referenced By	Gpt_HL_GetTimeElapsed()	

2.8.5.5 void Gpt_LL_Rtc_Init (void)

Function	void Gpt_LL_Rtc_Init (void)	
Description	RTC initialization	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_Init()	

2.8.5.6 void Gpt_LL_Rtc_StartTimer (uint32 u32Value)

Function	void Gpt_LL_Rtc_StartTimer (uint32 u32Value)	
Description	RTC start timer.	
Parameters	Parameter	Description
	u32Value	Start time value
Returns	N/A	
Referenced By	Gpt_HL_StartTimer()	

2.8.5.7 void Gpt_LL_Rtc_StopTimer (void)

Function	void Gpt_LL_Rtc_StopTimer (void)	
Description	RTC stop timer.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_HL_StopTimer()	

2.8.6 Functions in Gpt_Tstmp.h

2.8.6.1 void Gpt_LL_Tstmp_Init(Gpt_ModuleType eModule, uint8 u8Channel)

Function	void Gpt_LL_Tstmp_Init(Gpt_ModuleType eModule, uint8 u8Channel)	
Description	FTU initialization.	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
Returns	N/A	
Referenced By	Gpt_HL_Init()	

2.8.6.2

2.8.6.3 void Gpt_LL_Tstmp_DeInit(Gpt_ModuleType eModule, uint8 u8Channel)

Function	void Gpt_LL_Tstmp_DeInit(Gpt_ModuleType eModule, uint8 u8Channel)	
Description	De-initialization	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
Returns	N/A	
Referenced By	Gpt_HL_Init()	

2.8.6.4 void Gpt_LL_Tstmp_StartTimer(Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32Value)

Function	Gpt_LL_Tstmp_StartTimer(Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32Value)	
Description	TSTMP start timer.	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
	u32Value	Start timer value
Returns	N/A	
Referenced By	Gpt_HL_StartTimer()	

2.8.6.5 void Gpt_LL_Tstmp_StopTimer(Gpt_ModuleType eModule, uint8 u8Channel)

Function	Gpt_LL_Tstmp_StopTimer(Gpt_ModuleType eModule, uint8 u8Channel)	
Description	TSTMP stop timer.	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
Returns	N/A	
Referenced By	Gpt_HL_StartTimer()	

2.8.6.6 uint32 Gpt_LL_Tstmp_GetTimeElapsed(Gpt_ModuleType eModule, uint8 u8Channel, boolean *pbReturnChannelRollover, uint32 *pu32TargetValue)

Function	uint32 Gpt_LL_Tstmp_GetTimeElapsed(Gpt_ModuleType eModule, uint8 u8Channel, boolean *pbReturnChannelRollover, uint32 *pu32TargetValue)	
Description	TSTMP get elapsed value	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
	pReturnChannelRollover	RTC channel roll-over flag
	pTargetValue	Timer target value
Returns	uint32 Elapsed value	
Referenced By	Gpt_HL_GetTimeElapsed()	

2.8.6.7 void Gpt_LL_Tstmp_StartPredefTimer(Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32MaxValue)

Function	void Gpt_LL_Tstmp_StartPredefTimer(Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32MaxValue)	
Description	Start the predef timer.	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
	u32MaxValue	TSTMP MOD counter max value
Returns	N/A	
Referenced By	Gpt_HL_StartPredefTimer()	

2.8.6.8 uint32 Gpt_LL_Tstmp_GetPredefTimerValue(Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32MaxCount)

Function	uint32 Gpt_LL_Tstmp_GetPredefTimerValue(Gpt_ModuleType eModule, uint8 u8Channel, uint32 u32MaxCount)	
Description	Get the predef timer value.	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
	u32MaxValue	TSTMP MOD counter max value
Returns	uint32 Predef timer value	
Referenced By	Gpt_HL_GetPredefTimerValue()	

2.8.6.9 void Gpt_LL_Tstmp_StopPredefTimer(Gpt_ModuleType eModule, uint8 u8Channel)

Function	void Gpt_LL_Tstmp_StopPredefTimer(Gpt_ModuleType eModule, uint8 u8Channel)	
Description	TSTMP stop predef timer	
Parameters	Parameter	Description
	eModule	TSTMP module.
	u8Channel	TSTMP channel for the current module.
Returns	N/A	
Referenced By	Gpt_HL_StopPredefTimer()	

2.9 Peripheral Functions

2.9.1 Functions in Gpt_Aontiemr_RegOps.h

2.9.1.1 LOCAL_INLINE uint32 AONTIMER_HWA_CheckIntEnableState (void)

Function	LOCAL_INLINE uint32 AONTIMER_HWA_CheckIntEnableState (void)
Description	Check the AONTIMER interrupt enable state.
Parameters	N/A
Returns	AONTIMER interrupt enable state
Referenced By	Gpt_LL_Aontimer_EnableInterrupt()

2.9.1.2 LOCAL_INLINE uint32 AONTIMER_HWA_CheckIntFlag (void)

Function	LOCAL_INLINE uint32 AONTIMER_HWA_CheckIntFlag (void)
Description	Check the interrupt flag.
Parameters	N/A
Returns	AONTIMER interrupt flag
Referenced By	Gpt_LL_Aontimer_GetTimeElapsed()

2.9.1.3 LOCAL_INLINE boolean AONTIMER_HWA_CheckIntFlagAndEnable (void)

Function	LOCAL_INLINE boolean AONTIMER_HWA_CheckIntFlagAndEnable (void)
Description	Check the interrupt enable and interrupt flag.
Parameters	N/A
Returns	If the interrupt is enabled and interrupt flag is 1, return True; else return False.
Referenced By	Gpt_LL_Aontimer_Init()

2.9.1.4 LOCAL_INLINE void AONTIMER_HWA_ClearInterruptFlag (void)

Function	LOCAL_INLINE void AONTIMER_HWA_ClearInterruptFlag (void)
Description	Clear AONTIMER interrupt flags.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_DeInit(), Gpt_LL_Aontimer_DisableInterrupt(), Gpt_LL_Aontimer_EnableInterrupt(), Gpt_LL_Aontimer_Init(), Gpt_LL_Aontimer_StopPredefTimer(), and Gpt_LL_Aontimer_StopTimer()

2.9.1.5 LOCAL_INLINE void AONTIMER_HWA_ConfigModulePrescale (uint32 u32RegValue)

Function	LOCAL_INLINE void AONTIMER_HWA_ConfigModulePrescale (uint32 u32RegValue)	
Description	Configure the AONTIMER module prescale.	
Parameters	Parameter	Description
	u32RegValue	PSR register value
Returns	N/A	
Referenced By	Gpt_LL_Aontimer_DeInit()	

2.9.1.6 LOCAL_INLINE void AONTIMER_HWA_DisableBypassMode (void)

Function	LOCAL_INLINE void AONTIMER_HWA_DisableBypassMode (void)
Description	If disable bypass mode, the timer will enable the prescaler in timer counter mode or glitch filter in pulse mode.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_StartPredefTimer()

2.9.1.7 LOCAL_INLINE void AONTIMER_HWA_DisableModuleInterrupt (void)

Function	LOCAL_INLINE void AONTIMER_HWA_DisableModuleInterrupt (void)
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Description	Disable an AONTIMER module interrupt.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_DeInit(), Gpt_LL_Aontimer_DisableInterrupt(), Gpt_LL_Aontimer_Init(), Gpt_LL_Aontimer_StartPredefTimer(), and Gpt_LL_Aontimer_StopPredefTimer()

2.9.1.8 LOCAL_INLINE void AONTIMER_HWA_DisablePulseMode (void)

Function	LOCAL_INLINE void AONTIMER_HWA_DisablePulseMode (void)
Description	Disable the AONTIMER module pulse mode.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_Init(), and Gpt_LL_Aontimer_StartPredefTimer()

2.9.1.9 LOCAL_INLINE void AONTIMER_HWA_DisableTimer (void)

Function	LOCAL_INLINE void AONTIMER_HWA_DisableTimer (void)
Description	Disable the AONTIMER module timer.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_DeInit(), Gpt_LL_Aontimer_Init(), Gpt_LL_Aontimer_StartPredefTimer(), Gpt_LL_Aontimer_StartTimer(), Gpt_LL_Aontimer_StopPredefTimer(), and Gpt_LL_Aontimer_StopTimer()

2.9.1.10 LOCAL_INLINE void AONTIMER_HWA_EnableBypassMode (void)

Function	LOCAL_INLINE void AONTIMER_HWA_EnableBypassMode (void)
Description	If enable bypass mode, the timer will bypass the prescaler in timer counter mode or glitch filter in pulse mode.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_Init(), and Gpt_LL_Aontimer_StartPredefTimer()

2.9.1.11 LOCAL_INLINE void AONTIMER_HWA_EnableModuleInterrupt (void)

Function	LOCAL_INLINE void AONTIMER_HWA_EnableModuleInterrupt (void)
Description	Enable an AONTIMER module interrupt.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_EnableInterrupt().

2.9.1.12 LOCAL_INLINE void AONTIMER_HWA_EnableTimer (void)

Function	LOCAL_INLINE void AONTIMER_HWA_EnableTimer (void)
Description	Enable the AONTIMER timer.
Parameters	N/A
Returns	N/A

Referenced By	Gpt_LL_Aontimer_StartPredefTimer(), and Gpt_LL_Aontimer_StartTimer()
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2.9.1.13 LOCAL_INLINE uint32 AONTIMER_HWA_ReadCompareValue (void)

Function	LOCAL_INLINE uint32 AONTIMER_HWA_ReadCompareValue (void)
Description	Read the AONTIMER compare value.
Parameters	N/A
Returns	N/A AONTIMER compare value
Referenced By	Gpt_LL_Aontimer_GetTimeElapsed()

2.9.1.14 LOCAL_INLINE uint32 AONTIMER_HWA_ReadCounterCurrentValue (void)

Function	LOCAL_INLINE uint32 AONTIMER_HWA_ReadCounterCurrentValue (void)
Description	Read the AONTIMER current counter value.
Parameters	N/A
Returns	AONTIMER current counter value
Referenced By	Gpt_LL_Aontimer_GetPredefTimerValue(), and Gpt_LL_Aontimer_GetTimeElapsed()

2.9.1.15 LOCAL_INLINE void AONTIMER_HWA_SelectModuleClkSrc (Gpt_AontimerClkSrcType eClk)

Function	LOCAL_INLINE void AONTIMER_HWA_SelectModuleClkSrc (Gpt_AontimerClkSrcType eClk)	
Description	Select the AONTIMER module clock source.	
Parameters	Parameter	Description
	eClk	AONTIMER clock source
Returns	N/A	
Referenced By	Gpt_LL_Aontimer_Init(), and Gpt_LL_Aontimer_StartPredefTimer()	

2.9.1.16 LOCAL_INLINE void AONTIMER_HWA_SetCounterResetOnOverflow (void)

Function	LOCAL_INLINE void AONTIMER_HWA_SetCounterResetOnOverflow (void)
Description	Configure the CNR register to reset on overflow.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Aontimer_StartPredefTimer()

2.9.1.17 LOCAL_INLINE void AONTIMER_HWA_SetModuleCompareValue (uint32 u32RegValue)

Function	LOCAL_INLINE void AONTIMER_HWA_SetModuleCompareValue (uint32 u32RegValue)	
Description	Set the AONTIMER compare value.	
Parameters	Parameter	Description
	u32RegValue	CMR register value
Returns	N/A	
Referenced By	Gpt_LL_Aontimer_Init(), Gpt_LL_Aontimer_StartPredefTimer(), and Gpt_LL_Aontimer_StartTimer()	

2.9.1.18 LOCAL_INLINE void AONTIMER_HWA_SetModuleCounterValue (uint32 u32RegValue)

Function	LOCAL_INLINE void AONTIMER_HWA_SetModuleCounterValue (uint32 u32RegValue)
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Description	Set the AONTIMER current counter value.	
Parameters	Parameter	Description
	u32RegValue	CNR register value
Returns	N/A	
Referenced By	Gpt_LL_Aontimer_GetPredefTimerValue(), and Gpt_LL_Aontimer_GetTimeElapsed()	

2.9.1.19 LOCAL_INLINE void AONTIMER_HWA_SetPrescale (Gpt_AontimerPrescalerType ePrescalerValue)

Function	LOCAL_INLINE void AONTIMER_HWA_SetPrescale (Gpt_AontimerPrescalerType ePrescalerValue)	
Description	Set the AONTIMER prescaler.	
Parameters	Parameter	Description
	ePrescalerValue	Prescaler value,the range of the input value is :0~15, and the range of prescaler is :2^1 ~ 2^16.
Returns	N/A	
Referenced By	Gpt_LL_Aontimer_Init(), and Gpt_LL_Aontimer_StartPredefTimer()	

2.9.2 Functions in Gpt_Fcpit_RegOps.h

2.9.2.1 LOCAL_INLINE void FCPIT_HWA_ClearChannelsInterruptFlag (uint32 u32RegValue)

Function	LOCAL_INLINE void FCPIT_HWA_ClearChannelsInterruptFlag (uint32 u32RegValue)	
Description	Clear the FCPIT channel(n) interrupt flag.	
Parameters	Parameter	Description
	u32RegValue	0-3 bit indicates TIF0-TIF3
Returns	N/A	
Referenced By	Gpt_LL_FCPit_DeInit(), Gpt_LL_FCPIT_DisableInterrupt(), Gpt_LL_FCPit_EnableInterrupt(), Gpt_LL_Fcpit_Init(), Gpt_LL_FCPIT_StartPredefTimer(), and Gpt_LL_FCPit_StopTimer()	

2.9.2.2 LOCAL_INLINE void FCPIT_HWA_ConfigChannel (Gpt_FcpitChannelType eChannel, uint32 u32RegValue)

Function	LOCAL_INLINE void FCPIT_HWA_ConfigChannel (Gpt_FcpitChannelType eChannel, uint32 u32RegValue)	
Description	Configure the FCPIT channel.	
Parameters	Parameter	Description
	eChannel	FCPIT channel number
	u32RegValue	TCTRL register value
Returns	N/A	
Referenced By	Gpt_LL_FCPit_DeInit(), Gpt_LL_Fcpit_Init(), and Gpt_LL_FCPIT_StartPredefTimer()	

2.9.2.3 LOCAL_INLINE void FCPIT_HWA_DisableChannel (Gpt_FcpitChannelType eChannel)

Function	LOCAL_INLINE void FCPIT_HWA_DisableChannel (Gpt_FcpitChannelType eChannel)	
Description	Disable the FCPIT channel.	
Parameters	Parameter	Description
	eChannel	FCPIT channel number
Returns	N/A	
Referenced By	Gpt_LL_Fcpit_Init(), Gpt_LL_FCPIT_StartPredefTimer(), and Gpt_LL_FCPit_StartTimer(),	

	Gpt_LL_FCPIT_StopPredefTimer(), and Gpt_LL_FCPit_StopTimer()
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2.9.2.4 LOCAL_INLINE void FCPIT_HWA_DisableChannelsInterrupt (uint32 *u32RegValue*)

Function	LOCAL_INLINE void FCPIT_HWA_DisableChannelsInterrupt (uint32 <i>u32RegValue</i>)	
Description	Disable FCPIT channel(n) interrupt.	
Parameters	Parameter	Description
	u32RegValue	u32RegValue 0-3 bit indicate TIE0-TIE3
Returns	N/A	
Referenced By	Gpt_LL_FCPit_DeInit(), Gpt_LL_FCPIT_DisableInterrupt(), Gpt_LL_Fcpit_Init(), and Gpt_LL_FCPIT_StartPredefTimer()	

2.9.2.5 LOCAL_INLINE void FCPIT_HWA_DisableModule (void)

Function	LOCAL_INLINE void FCPIT_HWA_DisableModule (void)	
Description	Disable FCPIT module	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_LL_FCPit_DeInit()	

2.9.2.6 LOCAL_INLINE void FCPIT_HWA_EnableChannel (Gpt_FcpitChannelType *eChannel*)

Function	LOCAL_INLINE void FCPIT_HWA_EnableChannel (Gpt_FcpitChannelType <i>eChannel</i>)	
Description	Enables the FCPIT channel.	
Parameters	Parameter	Description
	eChannel	FCPIT channel number
Returns	N/A	
Referenced By	Gpt_LL_FCPIT_StartPredefTimer(), and Gpt_LL_FCPit_StartTimer()	

2.9.2.7 LOCAL_INLINE void FCPIT_HWA_EnableChannelsInterrupt (uint32 *u32RegValue*)

Function	LOCAL_INLINE void FCPIT_HWA_EnableChannelsInterrupt (uint32 <i>u32RegValue</i>)	
Description	Enable FCPIT channel(n) interrupt.	
Parameters	Parameter	Description
	u32RegValue	u32RegValue 0-3 bit indicates TIE0-TIE3
Returns	N/A	
Referenced By	Gpt_LL_FCPit_EnableInterrupt()	

2.9.2.8 LOCAL_INLINE void FCPIT_HWA_EnableModule (void)

Function	LOCAL_INLINE void FCPIT_HWA_EnableModule (void)	
Description	Enable the FCPIT module.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_LL_FCPit_DeInit(), Gpt_LL_Fcpit_Init(), and Gpt_LL_FCPIT_StartPredefTimer().	

2.9.2.9 LOCAL_INLINE uint32 FCPIT_HWA_ReadChannelTimerValue (Gpt_FcpitChannelType *eChannel*)

Function	LOCAL_INLINE uint32 FCPIT_HWA_ReadChannelTimerValue (Gpt_FcpitChannelType <i>eChannel</i>)	
Description	Read the FCPIT channel timer value.	
Parameters	Parameter	Description
	eChannel	FCPIT channel number
Returns	FCPIT channel timer value.	
Referenced By	Gpt_LL_FCPIT_GetPredefTimerValue(), and Gpt_LL_FCPit_GetTimeElapsed()	

2.9.2.10 LOCAL_INLINE uint32 FCPIT_HWA_ReadCurChannelValue (Gpt_FcpitChannelType *eChannel*)

Function	LOCAL_INLINE uint32 FCPIT_HWA_ReadCurChannelValue (Gpt_FcpitChannelType <i>eChannel</i>)	
Description	Read the current FCPIT channel value.	
Parameters	N/A	
Returns	FCPIT current timer value.	
Referenced By	Gpt_LL_FCPit_GetTimeElapsed()	

2.9.2.11 LOCAL_INLINE uint32 FCPIT_HWA_ReadEnableInterruptFlag (void)

Function	LOCAL_INLINE uint32 FCPIT_HWA_ReadEnableInterruptFlag (void)	
Description	Read the FCPIT enable interrupt flag.	
Parameters	N/A	
Returns	FCPIT enable interrupt flag	
Referenced By	ISR function	

2.9.2.12 LOCAL_INLINE uint32 FCPIT_HWA_ReadInterruptFlag (void)

Function	LOCAL_INLINE uint32 FCPIT_HWA_ReadInterruptFlag (void)	
Description	Read the FCPIT interrupt flag.	
Parameters	N/A	
Returns	FCPIT interrupt flag	
Referenced By	Gpt_LL_FCPit_GetTimeElapsed()	

2.9.2.13 LOCAL_INLINE void FCPIT_HWA_SetChannelRunOnDebug (void)

Function	LOCAL_INLINE void FCPIT_HWA_SetChannelRunOnDebug (void)	
Description	Set the FCPIT channel running on debug mode.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_LL_Fcpit_Init(), and Gpt_LL_FCPIT_StartPredefTimer()	

2.9.2.14 LOCAL_INLINE void FCPIT_HWA_SetChannelRunOnLpm (void)

Function	LOCAL_INLINE void FCPIT_HWA_SetChannelRunOnLpm (void)	
Description	Set the FCPIT channel running on low power mode.	
Parameters	N/A	
Returns	N/A	

Referenced By	Gpt_LL_Fcpit_Init()
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2.9.2.15 LOCAL_INLINE void FCPIT_HWA_SetChannelStopOnDebug (void)

Function	LOCAL_INLINE void FCPIT_HWA_SetChannelStopOnDebug (void)
Description	Set the FCPIT channel stop on debug mode.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_FCPit_DeInit(), Gpt_LL_Fcpit_Init(), and Gpt_LL_FCPIT_StartPredefTimer()

2.9.2.16 LOCAL_INLINE void FCPIT_HWA_SetChannelStopOnLpm (void)

Function	LOCAL_INLINE void FCPIT_HWA_SetChannelStopOnLpm (void)
Description	Set the FCPIT channel stop on low power mode.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_FCPit_DeInit()

2.9.2.17 LOCAL_INLINE void FCPIT_HWA_SetChannelValue (Gpt_FcpitChannelType *eChannel*, uint32 *u32RegValue*)

Function	LOCAL_INLINE void FCPIT_HWA_SetChannelValue (Gpt_FcpitChannelType <i>eChannel</i> , uint32 <i>u32RegValue</i>)	
Description	Set the FCPIT channel value.	
Parameters	Parameter	Description
	eChannel	FCPIT channel number .
	u32RegValue	Timer value.
Returns	N/A	
Referenced By	Gpt_LL_Fcpit_Init(), Gpt_LL_FCPIT_StartPredefTimer(), and Gpt_LL_FCPit_StartTimer()	

2.9.3 Functions in Gpt_Ftu_RegOps.h

2.9.3.1 LOCAL_INLINE void FTU_HWA_ClearChannelInterruptFlag (FTU_Type * *pFtu*, uint8 *u8Channel*)

Function	LOCAL_INLINE void FTU_HWA_ClearChannelInterruptFlag (FTU_Type * <i>pFtu</i> , uint8 <i>u8Channel</i>)	
Description	Clear the FTU channel interrupt flag.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number, range is 0-7.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DeInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), Gpt_LL_Ftu_StartTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.2 LOCAL_INLINE void FTU_HWA_ClearModuleAllFaultFlag (FTU_Type * *pFtu*)

Function	LOCAL_INLINE void FTU_HWA_ClearModuleAllFaultFlag (FTU_Type * <i>pFtu</i>)	
Description	Clear the FTU module all fault flag.	
Parameters	Parameter	Description

	pFtu	FTU instance.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_Init(), and Gpt_LL_Ftu_StartPredefTimer()	

2.9.3.3 LOCAL_INLINE void FTU_HWA_ClearModuleCounter (FTU_Type * pFtu, uint32 u32RegValue)

Function	LOCAL_INLINE void FTU_HWA_ClearModuleCounter (FTU_Type * pFtu, uint32 u32RegValue)	
Description	Clears the FTU counter.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u32RegValue	CNT register value.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DelInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.4 LOCAL_INLINE void FTU_HWA_ClearModuleCounterInitialValue (FTU_Type * pFtu, uint32 u32RegValue)

Function	LOCAL_INLINE void FTU_HWA_ClearModuleCounterInitialValue (FTU_Type * pFtu, uint32 u32RegValue)	
Description	Clear the FTU counter initial value.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u32RegValue	CNTIN register value.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DelInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.5 LOCAL_INLINE void FTU_HWA_ClearModuleUpdateRegBySync (FTU_Type * pFtu)

Function	LOCAL_INLINE void FTU_HWA_ClearModuleUpdateRegBySync (FTU_Type * pFtu)	
Description	Clear MODE[FTUEN]. This field defines different free-running counters and synchronization behaviors.	
Parameters	Parameter	Description
	pFtu	FTU instance.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DelInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.6 LOCAL_INLINE void FTU_HWA_ClearOutputMode (FTU_Type * pFtu, uint8 u8Channel)

Function	LOCAL_INLINE void FTU_HWA_ClearOutputMode (FTU_Type * pFtu, uint8 u8Channel)	
Description	Clear the channel output mode.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number, range is 0-7.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DelInit()	

2.9.3.7 LOCAL_INLINE void FTU_HWA_ConfigModuleDebugMode (FTU_Type * *pFtu*, Gpt_FtuDebugModeType *eDbgMode*)

Function	LOCAL_INLINE void FTU_HWA_ConfigModuleDebugMode (FTU_Type * <i>pFtu</i> , Gpt_FtuDebugModeType <i>eDbgMode</i>)	
Description	Configure the FTU module debug mode.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	eDbgMode	Debug mode enumeration.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DeInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.8 LOCAL_INLINE void FTU_HWA_DisableChannelInterrupt (FTU_Type * *pFtu*, uint8 *u8Channel*)

Function	LOCAL_INLINE void FTU_HWA_DisableChannelInterrupt (FTU_Type * <i>pFtu</i> , uint8 <i>u8Channel</i>)	
Description	Disable a channel interrupt.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number, range is 0-7.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DeInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), Gpt_LL_Ftu_StopPredefTimer(), and Gpt_LL_Ftu_StopTimer()	

2.9.3.9 LOCAL_INLINE void FTU_HWA_DisableModuleCpwmMode (FTU_Type * *pFtu*)

Function	LOCAL_INLINE void FTU_HWA_DisableModuleCpwmMode (FTU_Type * <i>pFtu</i>)	
Description	Disable the FTU module CPWM mode.	
Parameters	Parameter	Description
	pFtu	FTU instance.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_StartPredefTimer()	

2.9.3.10 LOCAL_INLINE void FTU_HWA_DisableModuleFaultMode (FTU_Type * *pFtu*)

Function	LOCAL_INLINE void FTU_HWA_DisableModuleFaultMode (FTU_Type * <i>pFtu</i>)	
Description	Disable the FTU module fault mode.	
Parameters	Parameter	Description
	pFtu	FTU instance.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_Init(), and Gpt_LL_Ftu_StartPredefTimer()	

2.9.3.11 LOCAL_INLINE void FTU_HWA_DisableOverflowInterrupt (FTU_Type * *pFtu*)

Function	LOCAL_INLINE void FTU_HWA_DisableOverflowInterrupt (FTU_Type * <i>pFtu</i>)	
Description	Disable an FTU module overflow interrupt.	

Parameters	Parameter	Description
	pFtu	FTU instance.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_Init(), and Gpt_LL_Ftu_StartPredefTimer()	

2.9.3.12 LOCAL_INLINE void FTU_HWA_DisableQuadratureMode (FTU_Type * pFtu)

Function	LOCAL_INLINE void FTU_HWA_DisableQuadratureMode (FTU_Type * pFtu)	
Description	Disable the FTU module quadrature mode.	
Parameters	Parameter	Description
	pFtu	FTU instance.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_StartPredefTimer()	

2.9.3.13 LOCAL_INLINE void FTU_HWA_EnableChannelInterrupt (FTU_Type * pFtu, uint8 u8Channel)

Function	LOCAL_INLINE void FTU_HWA_EnableChannelInterrupt (FTU_Type * pFtu, uint8 u8Channel)	
Description	Enable an FTU channel interrupt.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number, range is 0-7.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_StartTimer()	

2.9.3.14 LOCAL_INLINE uint32 FTU_HWA_ReadChannelInterruptFlag (FTU_Type * pFtu, uint8 u8Channel)

Function	LOCAL_INLINE uint32 FTU_HWA_ReadChannelInterruptFlag (FTU_Type * pFtu, uint8 u8Channel)	
Description	Read the FTU channel interrupt flag.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number, range is 0-7.
Returns	FTU channel interrupt flag	
Referenced By	Gpt_LL_Ftu_GetTimeElapsed(), and Gpt_LL_Ftu_StartTimer()	

2.9.3.15 LOCAL_INLINE boolean FTU_HWA_ReadChannelInterruptFlagAndEnable (FTU_Type * pFtu, uint8 u8Channel)

Function	LOCAL_INLINE boolean FTU_HWA_ReadChannelInterruptFlagAndEnable (FTU_Type * pFtu, uint8 u8Channel)	
Description	Check the FTU channel interrupt flag and interrupt enable.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number.
Returns	If the channel interrupt flag exists and the interrupt is enabled, return true; else return false.	
Referenced By	ISR function	

2.9.3.16 LOCAL_INLINE uint32 FTU_HWA_ReadChannelValue (FTU_Type * *pFtu*, uint8 *u8Channel*)

Function	LOCAL_INLINE uint32 FTU_HWA_ReadChannelValue (FTU_Type * <i>pFtu</i> , uint8 <i>u8Channel</i>)	
Description	Read the FTU instance channel value.	
Parameters	Parameter	Description
	<i>pFtu</i>	FTU instance.
	<i>u8Channel</i>	FTU channel.
Returns	FTU channel value.	
Referenced By	Gpt_LL_Ftu_GetTimeElapsed().	

2.9.3.17 LOCAL_INLINE uint32 FTU_HWA_ReadModuleCounter (FTU_Type * *pFtu*)

Function	LOCAL_INLINE uint32 FTU_HWA_ReadModuleCounter (FTU_Type * <i>pFtu</i>)	
Description	Read the FTU module counter value.	
Parameters	Parameter	Description
	<i>pFtu</i>	FTU instance.
Returns	FTU module counter value.	
Referenced By	Gpt_LL_Ftu_GetPredefTimerValue(), Gpt_LL_Ftu_GetTimeElapsed(), and Gpt_LL_Ftu_StartTimer()	

2.9.3.18 LOCAL_INLINE void FTU_HWA_SetChannelValue (FTU_Type * *pFtu*, uint8 *u8Channel*, uint32 *u32RegValue*)

Function	LOCAL_INLINE void FTU_HWA_SetChannelValue (FTU_Type * <i>pFtu</i> , uint8 <i>u8Channel</i> , uint32 <i>u32RegValue</i>)	
Description	Set the FTU channel value.	
Parameters	Parameter	Description
	<i>pFtu</i>	FTU instance.
	<i>u8Channel</i>	Channel number, range is 0-7.
	<i>u32RegValue</i>	CnV register value.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DeInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), Gpt_LL_Ftu_StartTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.19 LOCAL_INLINE void FTU_HWA_SetModuleClkSrc (FTU_Type * *pFtu*, Gpt_FtuModuleClkSrcType *eClkSrc*)

Function	LOCAL_INLINE void FTU_HWA_SetModuleClkSrc (FTU_Type * <i>pFtu</i> , Gpt_FtuModuleClkSrcType <i>eClkSrc</i>)	
Description	Set the FTU module clock source.	
Parameters	Parameter	Description
	<i>pFtu</i>	FTU instance.
	<i>eClkSrc</i>	FTU module clock source type.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DeInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), and Gpt_LL_Ftu_StopPredefTimer()	

2.9.3.20 LOCAL_INLINE void FTU_HWA_SetModuleCompareValue (FTU_Type * *pFtu*, uint32 *u32RegValue*)

Function	LOCAL_INLINE void FTU_HWA_SetModuleCompareValue (FTU_Type * <i>pFtu</i> , uint32 <i>u32RegValue</i>)
-----------------	---

Description	Set the FTU counter compare value.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u32RegValue	MOD register value.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_DeInit(), Gpt_LL_Ftu_Init(), Gpt_LL_Ftu_StartPredefTimer(), and Gpt_LL_Ftu_StopPredefTimer().	

2.9.3.21 LOCAL_INLINE void FTU_HWA_SetModulePrescale (FTU_Type * pFtu, Gpt_FtuPrescalerType ePs)

Function	LOCAL_INLINE void FTU_HWA_SetModulePrescale (FTU_Type * pFtu, Gpt_FtuPrescalerType ePs)	
Description	Set the FTU module prescale	
Parameters	Parameter	Description
	pFtu	FTU instance.
	ePs	FTU clock prescaler enumeration.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_Init(), and Gpt_LL_Ftu_StartPredefTimer()	

2.9.3.22 LOCAL_INLINE void FTU_HWA_SetOutputMode (FTU_Type * pFtu, uint8 u8Channel)

Function	LOCAL_INLINE void FTU_HWA_SetOutputMode (FTU_Type * pFtu, uint8 u8Channel)	
Description	Configure the channel to output mode and signal not bind to pad.	
Parameters	Parameter	Description
	pFtu	FTU instance.
	u8Channel	FTU channel number, range is 0-7.
Returns	N/A	
Referenced By	Gpt_LL_Ftu_Init()	

2.9.4 Functions in Gpt_Rtc_RegOps.h

2.9.4.1 LOCAL_INLINE void ConfigControl (uint32 u32Value)

Function	LOCAL_INLINE void ConfigControl (uint32 u32Value)	
Description	Configure the control register.	
Parameters	Parameter	Description
	u32Value	Control value.
Returns	N/A	
Referenced By	Gpt_LL_Rtc_Init()	

2.9.4.2 LOCAL_INLINE void DisableAlarmInterrupt (void)

Function	LOCAL_INLINE void DisableAlarmInterrupt (void)	
Description	Disable an alarm interrupt.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_LL_Rtc_DisableInterrupt(), Gpt_LL_Rtc_Init(), Gpt_LL_Rtc_StartTimer(), and Gpt_LL_Rtc_StopTimer()	

2.9.4.3 LOCAL_INLINE void DisableOverflowInvalidInterrupt (void)

Function	LOCAL_INLINE void DisableOverflowInvalidInterrupt (void)
Description	Disable an overflow/invalid interrupt.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Rtc_Init()

2.9.4.4 LOCAL_INLINE void DisableRtcCounter (void)

Function	LOCAL_INLINE void DisableRtcCounter (void)
Description	Disable the RTC time counter.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Rtc_Init(), Gpt_LL_Rtc_StartTimer(), and Gpt_LL_Rtc_StopTimer()

2.9.4.5 LOCAL_INLINE void EnableAlarmInterrupt (void)

Function	LOCAL_INLINE void EnableAlarmInterrupt (void)
Description	Enable an alarm interrupt.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Rtc_EnableInterrupt(), and Gpt_LL_Rtc_StartTimer()

2.9.4.6 LOCAL_INLINE void EnableRtcCounter (void)

Function	LOCAL_INLINE void EnableRtcCounter (void)
Description	Enable the RTC time counter.
Parameters	N/A
Returns	N/A
Referenced By	Gpt_LL_Rtc_StartTimer()

2.9.4.7 LOCAL_INLINE uint32 ReadAlarmFlag (void)

Function	LOCAL_INLINE uint32 ReadAlarmFlag (void)
Description	Read the RTC alarm flag.
Parameters	N/A
Returns	Alarm flag
Referenced By	Gpt_LL_Rtc_GetTimeElapsed()

2.9.4.8 LOCAL_INLINE uint32 ReadAlarmInterruptEnableBit (void)

Function	LOCAL_INLINE uint32 ReadAlarmInterruptEnableBit (void)
Description	Read the RTC alarm interrupt enable bit.
Parameters	N/A
Returns	Alarm interrupt enable

Referenced By	Gpt_LL_Rtc_EnableInterrupt(), and Gpt_LL_Rtc_Init()
----------------------	---

2.9.4.9 LOCAL_INLINE uint32 ReadCounterEnableStatus (void)

Function	LOCAL_INLINE uint32 ReadCounterEnableStatus (void)	
Description	Read the RTC enable status.	
Parameters	N/A	
Returns	RTC enable status bit	
Referenced By	ISR function	

2.9.4.10 LOCAL_INLINE uint32 ReadIntFlag (void)

Function	LOCAL_INLINE uint32 ReadIntFlag (void)	
Description	Read the RTC status.	
Parameters	N/A	
Returns	RTC flag status	
Referenced By	ISR function	

2.9.4.11 LOCAL_INLINE uint32 ReadSecondValue (void)

Function	LOCAL_INLINE uint32 ReadSecondValue (void)	
Description	Read second value.	
Parameters	N/A	
Returns	Second value	
Referenced By	Gpt_LL_Rtc_GetTimeElapsed(), and Gpt_LL_Rtc_StartTimer()	

2.9.4.12 LOCAL_INLINE uint32 ReadTargetAlarmValue (void)

Function	LOCAL_INLINE uint32 ReadTargetAlarmValue (void)	
Description	Read the target alarm value.	
Parameters	N/A	
Returns	Target alarm value	
Referenced By	Gpt_LL_Rtc_DisableInterrupt(), Gpt_LL_Rtc_EnableInterrupt(), Gpt_LL_Rtc_GetTimeElapsed(), and Gpt_LL_Rtc_StopTimer()	

2.9.4.13 LOCAL_INLINE void RtcSoftwareReset (void)

Function	LOCAL_INLINE void RtcSoftwareReset (void)	
Description	RTC software reset.	
Parameters	N/A	
Returns	N/A	
Referenced By	Gpt_LL_Rtc_DeInit()	

2.9.4.14 LOCAL_INLINE void SetAlarmCounterValue (uint32 u32Value)

Function	LOCAL_INLINE void SetAlarmCounterValue (uint32 u32Value)	
Description	Set the RTC alarm value.	
Parameters	Parameter	Description

	u32Value	TAR register value.
Returns	N/A	
Referenced By	Gpt_LL_Rtc_DisableInterrupt(), Gpt_LL_Rtc_EnableInterrupt(), Gpt_LL_Rtc_StartTimer(), and Gpt_LL_Rtc_StopTimer()	

2.9.4.15 LOCAL_INLINE void SetPrescalerCounterValue (uint32 u32Value)

Function	LOCAL_INLINE void SetPrescalerCounterValue (uint32 u32Value)	
Description	Set the RTC prescaler register.	
Parameters	Parameter	Description
	u32Value	PR register value.
Returns	N/A	
Referenced By	Gpt_LL_Rtc_Init()	

2.9.4.16 LOCAL_INLINE void SetSecondCounterValue (uint32 u32Value)

Function	LOCAL_INLINE void SetSecondCounterValue (uint32 u32Value)	
Description	Set the RTC seconds register.	
Parameters	Parameter	Description
	u32Value	SR register value.
Returns	N/A	
Referenced By	Gpt_LL_Rtc_Init()	

2.9.5 Functions in Gpt_Tstmp_RegOps.h

2.9.5.1 LOCAL_INLINE uint64 TSTMP_HWA_ReadTstmpValue(TSTMP_Type *pTstmp)

Function	LOCAL_INLINE uint64 TSTMP_HWA_ReadTstmpValue(TSTMP_Type *pTstmp)	
Description	Read TSTMP value	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
Returns	TSTMP value	
Referenced By	Gpt_LL_Tstmp_StartTimer(), Gpt_LL_Tstmp_GetTimeElapsed(), Gpt_LL_Tstmp_StartPredefTimer(), Gpt_LL_Tstmp_GetPredefTimerValue()	

2.9.5.2 LOCAL_INLINE boolean TSTMP_HWA_ReadTstmpInterruptFlagAndEnable(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE boolean TSTMP_HWA_ReadTstmpInterruptFlagAndEnable(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Read TSTMP value	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	If channel interrupt flag exist and interrupt enable ,return TRUE, else return FALSE	
Referenced By	TSTMP_0_ISR(), TSTMP_1_ISR(), TSTMP_2_ISR(), TSTMP_3_ISR()	

2.9.5.3 LOCAL_INLINE void TSTMP_HWA_EnableModMatchInterrupt(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE void TSTMP_HWA_EnableModMatchInterrupt(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Enable TSTMP MOD(n) match interrupt	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_StartTimer()	

2.9.5.4 LOCAL_INLINE void TSTMP_HWA_DisableModMatchInterrupt(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE void TSTMP_HWA_DisableModMatchInterrupt(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Disable TSTMP MOD(n) match interrupt	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_Init(),Gpt_LL_Tstmp_DeInit(),Gpt_LL_Tstmp_StopTimer(),Gpt_LL_Tstmp_StartPredefTimer(),Gpt_LL_Tstmp_StopPredefTimer()	

2.9.5.5 LOCAL_INLINE void TSTMP_HWA_SetModCounterMode(TSTMP_Type *pTstmp, uint8 u8Mod, TSTMP_ModeCounterRunningMode eCounterMode)

Function	LOCAL_INLINE void TSTMP_HWA_SetModCounterMode(TSTMP_Type *pTstmp, uint8 u8Mod, TSTMP_ModeCounterRunningMode eCounterMode)	
Description	Set the counting modes of TSTMP MOD(n)	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
	eCounterMode	Counting mode set
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_Init(),Gpt_LL_Tstmp_DeInit(),Gpt_LL_Tstmp_StartPredefTimer(),Gpt_LL_Tstmp_StartPredefTimer(),Gpt_LL_Tstmp_StopPredefTimer()	

2.9.5.6 LOCAL_INLINE void TSTMP_HWA_EnableModCounter(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE void TSTMP_HWA_EnableModCounter(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Enable TSTMP MOD(n) counter	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_StartTimer(),Gpt_LL_Tstmp_StartPredefTimer()	

2.9.5.7 LOCAL_INLINE void TSTMP_HWA_DisableModCounter(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE void TSTMP_HWA_DisableModCounter(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Enable TSTMP MOD(n) counter	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_Init(),Gpt_LL_Tstmp_DeInit(),Gpt_LL_Tstmp_StopTimer(),Gpt_LL_Tstmp_StartPredefTimer(),Gpt_LL_Tstmp_StopPredefTimer()	

2.9.5.8 LOCAL_INLINE void TSTMP_HWA_SetModCounterMode(TSTMP_Type *pTstmp, uint8 u8Mod, TSTMP_ModeCounterRunningMode eCounterMode)

Function	LOCAL_INLINE void TSTMP_HWA_SetModCounterMode(TSTMP_Type *pTstmp, uint8 u8Mod, TSTMP_ModeCounterRunningMode eCounterMode)	
Description	Set the counting modes of TSTMP MOD(n)	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
	eModClkSrc	Clock source
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_Init(),Gpt_LL_Tstmp_DeInit()	

2.9.5.9 LOCAL_INLINE uint32 TSTMP_HWA_ReadModMatchFlag(TSTMP_Type *pTstmp)

Function	LOCAL_INLINE uint32 TSTMP_HWA_ReadModMatchFlag(TSTMP_Type *pTstmp)	
Description	Read TSTMP all MOD match flag	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
Returns	TSTMP value	
Referenced By	Gpt_LL_Tstmp_GetTimeElapsed(),TSTMP_0_ISR(),TSTMP_1_ISR(),TSTMP_2_ISR(), TSTMP_3_ISR()	

2.9.5.10 LOCAL_INLINE void TSTMP_HWA_ClearSingleMod123MatchFlag(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE void TSTMP_HWA_ClearSingleMod123MatchFlag(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Clear TSTMP single MOD match flag	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_Init(),Gpt_LL_Tstmp_DeInit(),Gpt_LL_Tstmp_StartPredefTimer(),Gpt_LL_Tstmp_StopPredefTimer(),Gpt_Tstmp_ProcessInterrupt(),TSTMP_0_ISR(),TSTMP_1_ISR(),TSTMP_2_ISR(), TSTMP_3_ISR()	

2.9.5.11 LOCAL_INLINE void TSTMP_HWA_SetModMatchValue(TSTMP_Type *pTstmp, uint8 u8Mod, uint32 u32ModValue)

Function	LOCAL_INLINE void TSTMP_HWA_SetModMatchValue(TSTMP_Type *pTstmp, uint8 u8Mod, uint32 u32ModValue)	
Description	Set MOD match value	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
	u32ModValue	MOD value
Returns	N/A	
Referenced By	Gpt_LL_Tstmp_DeInit(),Gpt_LL_Tstmp_StartTimer(),Gpt_LL_Tstmp_StartPredefTimer(),Gpt_LL_Tstmp_StopPredefTimer()	

2.9.5.12 LOCAL_INLINE uint32 TSTMP_HWA_GetModMatchValue(TSTMP_Type *pTstmp, uint8 u8Mod)

Function	LOCAL_INLINE uint32 TSTMP_HWA_GetModMatchValue(TSTMP_Type *pTstmp, uint8 u8Mod)	
Description	Get MOD match value	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
Returns	MOD match value	
Referenced By	Gpt_LL_Tstmp_GetTimeElapsed()	

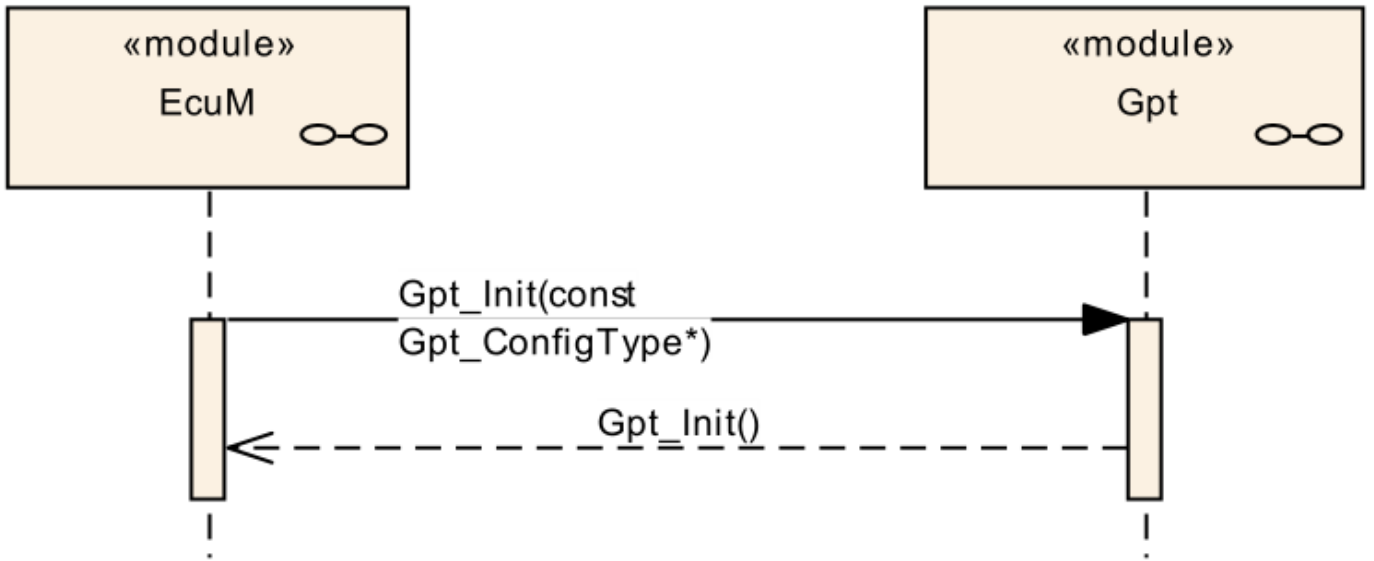
2.9.5.13 LOCAL_INLINE void TSTMP_HWA_SetNextTargetChannelValue(TSTMP_Type *pTstmp, uint8 u8Mod,uint64 *u64TargetValue)

Function	LOCAL_INLINE void TSTMP_HWA_SetNextTargetChannelValue(TSTMP_Type *pTstmp, uint8 u8Mod,uint64 *u64TargetValue)	
Description	Set next TSTMP MOD target value	
Parameters	Parameter	Description
	pTstmp	TSTMP instance
	u8Mod	MOD number
	u64TargetValue	MOD next target value
Returns	N/A	
Referenced By	Gpt_Tstmp_ProcessInterrupt()	

2.10 API Sequence Diagram

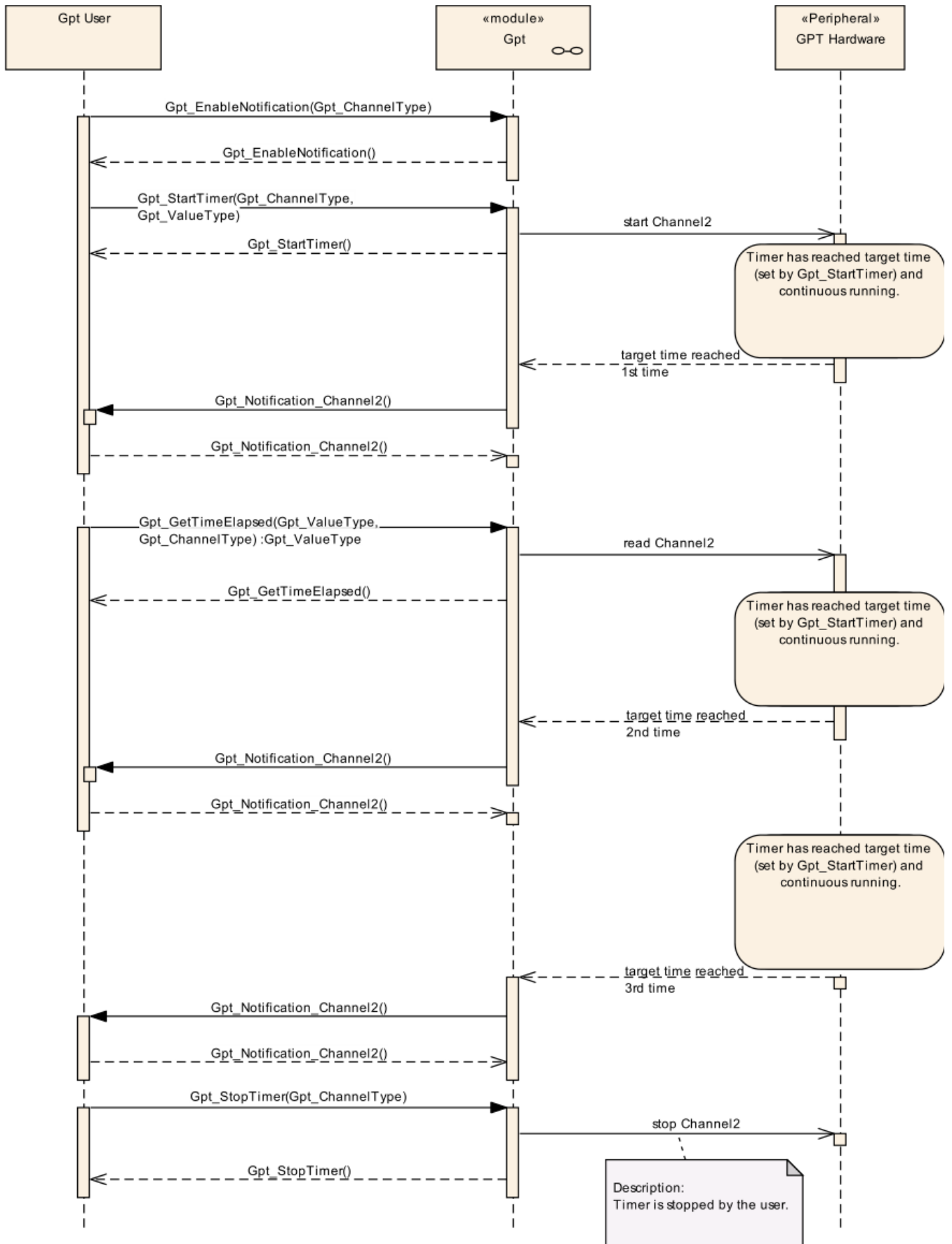
2.10.1 Gpt_Init

The ECU State Manager (EcuM) is responsible for calling the initialization function.



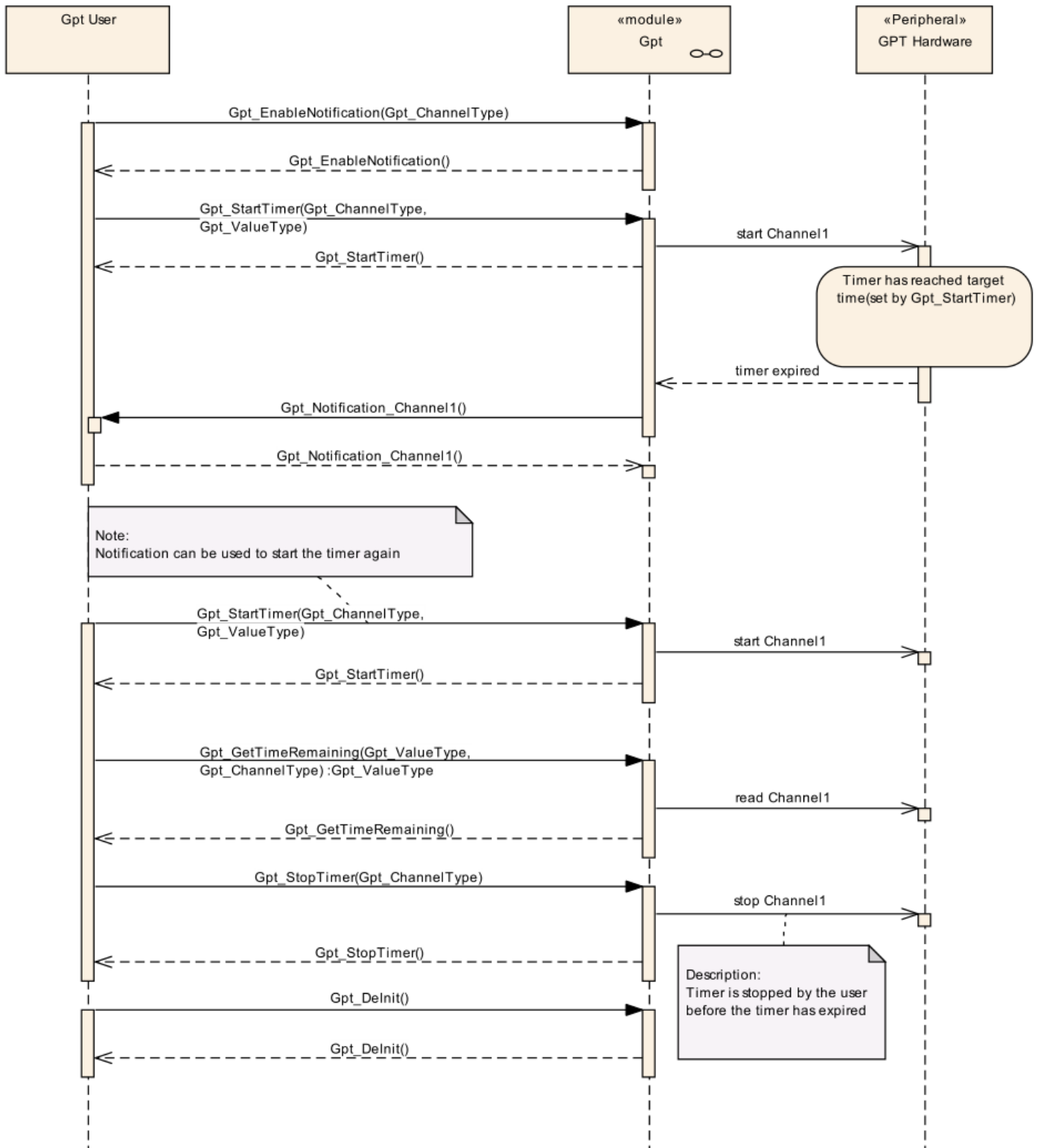
2.10.2 GPT Continuous Mode

Channel 2 is configured as "Continuous Mode."



2.10.3 GPT One-shot Mode

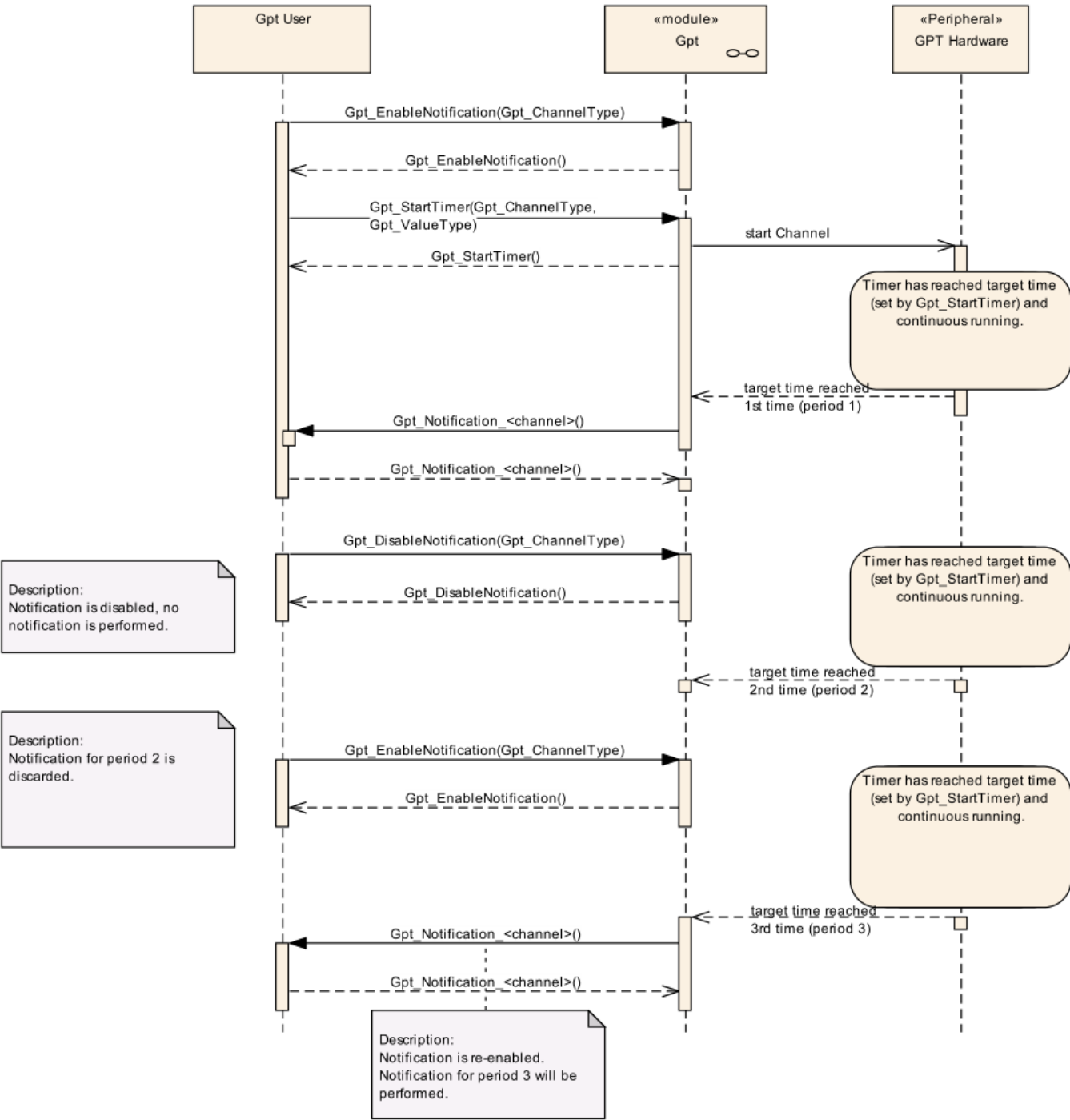
Channel 1 is configured for "One-shot Mode."



2.10.4 Disable/Enable Notifications

The sequence diagram shown in this chapter explains the behavior of the driver, when the notification is disabled, while the timer is still running in continuous mode. If the notification is disabled, the user will not be informed, when the timer reaches the target time the 2nd time (period 2).

This notification is discarded and not made up again, when the notification is re-enabled.



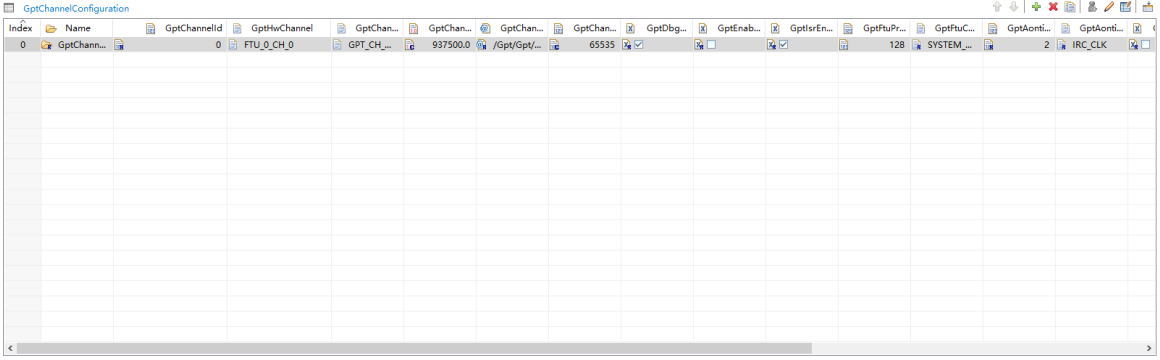
The container inclusion relation is shown below:



Container	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

Container	GptChannelConfigSet	
Description	This container is the base of a Configuration Set which contains the configured GPT channels. In this way, different configuration sets can be defined for the post-build process.	
Screenshot		
Properties	Property	Value
	Type	Container

Container	GptChannelConfiguration
Description	N/A

Screenshot		
Properties	Property	Value
	Type	List
	Min	1

3.2.2.1.1 GptChannelConfiguration

Container	GptChannelConfiguration	
Description	This container contains the channel-wide configuration (parameters) of the GPT Driver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1.2 GptChannelId

Variable	GptChannelId	
Description	Channel ID of the GPT channel. This value will be assigned to the symbolic name derived from the GptChannelConfiguration container short name.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	True
	Range	<= 4294967295, >=0

3.2.2.1.3 GptHwChannel

Variable	GptHwChannel	
Description	Select the physical GPT Channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Range	See ARM_fc7xxx.xxx.properties for details

3.2.2.1.4 GptChannelMode

Variable	GptChannelMode	
Description	Specify the behavior of the timer channel after the timeout has expired.	
Screenshot		
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	GPT_CH_MODE_ONESHOT

3.2.2.1.5 GptChannelTickFrequency

Variable	GptChannelTickFrequency	
Description	Specify the tick frequency of the timer channel in Hz	
Screenshot		
Properties	Property	Value
	Type	Variable: Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	See MCU configuration for details

3.2.2.1.6 GptChannelClkSrcRef

Variable	GptChannelClkSrcRef	
Description	Reference to the GptClockReferencePoint from which the channel clock is derived.	
Screenshot		
Properties	Property	Value
	Type	Variable: Reference
	Origin	AUTOSAR_ECUC
	Ref	Reference to node: GptClockReferencePoint

3.2.2.1.7 GptChannelTickValueMax

Variable	GptChannelTickValueMax	
Description	Maximum value in ticks, the timer channel is able to count. With the next tick, the timer rolls over to zero. There is mandatory to set 4294967295 for the FCPIT and RTC channels (corresponding to the 32 bits counter resolution) and 65535 for the FTU and AONTIMER channels (corresponding to the 16 bits counter resolution).	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	See MCU configuration for details

3.2.2.1.8 GptDbgModeEnable

Variable	GptDbgModeEnable	
Description	If enable, the counter will enable in debug mode. This feature just for FTU and AONTIMER hw resource. For FTU channels the debug mode option must be the same for all channels belonging to the same module.	
Screenshot	GptDbgModeEnable    	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.1.9 GptEnableWakeup

Variable	GptEnableWakeup	
Description	Enable wakeup capability of CPU for a channel.	
Screenshot	GptEnableWakeup    	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	false

3.2.2.1.10 GptIsrEnable

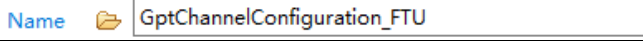
Variable	GptIsrEnable	
Description	Enable/Disable HW channels' Interrupt Sources. For FTU channels, the ISR enable option must be the same for all channels belonging to the same module.	
Screenshot	GptIsrEnable    	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.1.11 GptNotification

Variable	GptNotification	
Description	Function pointer to callback function (for non-wakeup notification). The field is editable only if the switch GptEnableDisableNotificationApi is true.	
Screenshot	 GptNotification  NULL_PTR  	
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	NULL_PTR

	Optional	true
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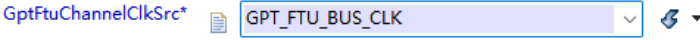
3.2.2.2 GptChannelConfiguration_FTU

Container	GptChannelConfiguration_FTU	
Description	This container contains the FTU channel-wide configuration (parameters) of the GPT Driver.	
Screenshot		
Properties	Property	Value
	Type	Container
	Frame	line

3.2.2.2.1 GptFtuPrescaler

Variable	GptFtuPrescaler	
Description	The GPT module specific clock prescaler value. Selects one of 8 division factors for the clock source selected by GptChannelClkSrcRef. The new prescaler factor affects the clock source on the next system clock cycle after the new value is updated into the register bits. For FTU channels the prescaler option must be the same for all channels belonging to the same module	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	1
	Range	1,2,4,8,16,32,64,128

3.2.2.2.2 GptFtuChannelClkSrc

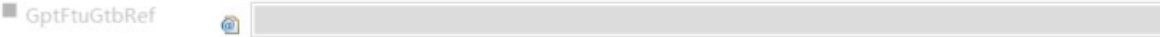
Variable	GptFtuChannelClkSrc	
Description	The GPT module specific clock input for the timer unit can statically be configured and allows to select different clock sources per module. For FTU channels the clock source option must be the same for all channels belonging to the same module	
Screenshot		
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	GPT_FTU_BUS_CLOCK
	Range	GPT_FTU_BUS_CLOCK, GPT_FTU_PCC_CLK, GPT_FTU_EXTERNAL_CLK0, GPT_FTU_EXTERNAL_CLK1, GPT_FTU_EXTERNAL_CLK2

3.2.2.2.3 GptFtuGtb

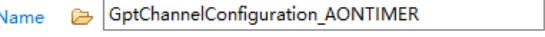
Variable	GptFtuGtb
Description	Switch for enable FTU Global Time Base.(Only available for FTU peripherals)

Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.2.4 GptFtuGtbRef

Variable	GptFtuGtbRef	
Description	EN: Maps the Gpt ftu instance to Gpt gtb group	
Screenshot		
Properties	Property	Value
	Type	Variable: Reference
	Origin	FLAGCHIP
	Ref	Reference to node: GptFtuGtbGroup

3.2.2.3 GptChannelConfiguration_AONTIMER

Container	GptChannelConfiguration_AONTIMER	
Description	This container contains the AONTIMER channel-wide configuration (parameters) of the GPT Driver.	
Screenshot		
Properties	Property	Value
	Type	Container
	Frame	line

3.2.2.3.1 GptAontimerPrescaler

Variable	GptAontimerPrescaler	
Description	The GPT module specific clock prescaler value. Configures the Prescaler in Time Counter mode, from one of 16 divisions.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	2
	Range	1,2,4,8,16,32,64,128,256,512,1024,2048,4096,8192,16384,32768,65536

3.2.2.3.2 GptAontimerChannelClkSrc

Variable	GptAontimerChannelClkSrc	
Description	The GPT module specific clock input for the timer unit can statically be configured and allows to select different clock sources per module. Select the clock source for the AONTIMER module for this platform	
Screenshot		

Properties	Property	Value
	Type	Variable: Enumeration
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	SIRC_1MHZ
	Range	SIRC_1MHZ,RTC_CLK,IRC_CLK

3.2.2.4 GptChannelConfiguration_FCPIT

Container	GptChannelConfiguration_FCPIT	
Description	This container contains the FCPIT channel-wide configuration (parameters) of the GPT Driver	
Screenshot		
Properties	Property	Value
	Type	Container
	Frame	line

3.2.2.4.1 GptFCPitIsExternalTrigger

Variable	GptFCPitIsExternalTrigger	
Description	Select between external/internal trigger sources.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.4.2 GptFCPitEnReloadOnTrigger

Variable	GptFCPitEnReloadOnTrigger	
Description	FCPIT Timer channel will Reload Timer at each rising edge of trigger source.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.4.3 GptFCPitEnStopOnInterrupt

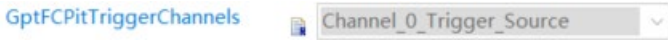
Variable	GptFCPitEnStopOnInterrupt	
Description	FCPIT Timer Channel will stop when its interrupt occurs.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean

	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.4.4 GptFCPitEnStartOnTrigger

Variable	GptFCPitEnStartOnTrigger	
Description	FCPIT Timer Channel will Start at each rising edge of Trigger Signal.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.2.4.5 GptFCPitTriggerChannels

Variable	GptFCPitTriggerChannels	
Description	FCPIT External Trigger Channels Select. Trigger Channels can be selected from 16 external sources. 0 Timer channel 0 trigger source is selectd. 1 Timer channel 1 trigger source is selectd. 14 Timer channel 14 trigger source is selectd. 15 Timer channel 15 trigger source is selectd.	
Screenshot		
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	Channel_0_Trigger_Source
	Range	Channel_0_Trigger_Source, Channel_1_Trigger_Source, Channel_2_Trigger_Source, Channel_3_Trigger_Source, Channel_4_Trigger_Source, Channel_5_Trigger_Source, Channel_6_Trigger_Source, Channel_7_Trigger_Source, Channel_8_Trigger_Source, Channel_9_Trigger_Source, Channel_10_Trigger_Source, Channel_11_Trigger_Source, Channel_12_Trigger_Source, Channel_13_Trigger_Source, Channel_14_Trigger_Source, Channel_15_Trigger_Source

3.2.2.4.6 GptFCPitChainMode

Variable	GptFCPitChainMode
Description	Enable/disable Chain Mode for FCPIT.

Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

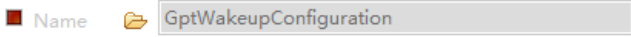
3.2.2.5 GptChannelConfiguration_TSTMP

Container	GptChannelConfiguration_TSTMP	
Description	The Clock Source of Modulate Timer in TSTMP0 can be selected in range of AON_CLK or SRIC_1M. Note: Available on FC7240 TSTMP0 only.	
Screenshot		
Properties	Property	Value
	Type	Container
	Frame	line

3.2.2.5.1 GptTstmpModClkSrcSelect

Variable	GptTstmpModClkSrcSelect	
Description	The Clock Source of Modulate Timer in TSTMP0 can be selected in range of AON_CLK or SRIC_1M. Note: Available on FC7240 TSTMP0 only.	
Screenshot		
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	SRIC_1M
	Range	SRIC_1M, AON_CLK

3.2.2.6 GptWakeupConfiguration

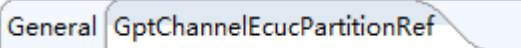
Container	GptWakeupConfiguration	
Description	This container defines the wakeup source codes reported to Ecu State Manager	
Screenshot		
Properties	Property	Value
	Type	Container
	Frame	line
	Optional	true
	Enable	false

3.2.2.6.1 GptWakeupSourceRef

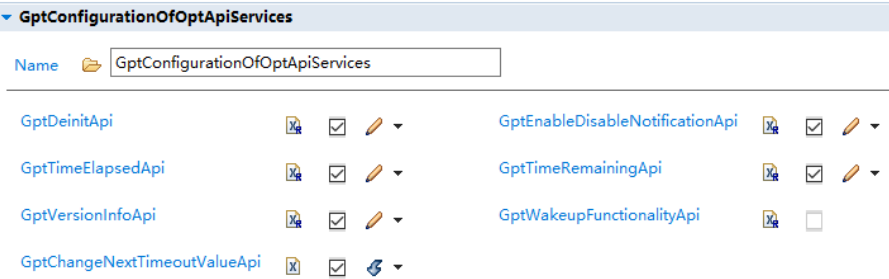
Variable	GptWakeupSourceRef
-----------------	--------------------

Description	In case the wakeup-capability is true this value is transmitted to the Ecu State Manager. Implementation Type: reference to EcuM_WakeupSourceType.	
Screenshot		
Properties	Property	Value
	Type	Variable: Reference
	Origin	AUTOSAR_ECUC
	Ref	Reference to node: EcuMWakeupSource

3.2.2.6.2 GptChannelEcucPartitionRef

Variable	GptChannelEcucPartitionRef	
Description	Maps a GPT channel to zero or multiple ECUC partitions to limit the access to this channel group. The ECUC partitions referenced are a subset of the ECUC partitions where the GPT driver is mapped to	
Screenshot		
Properties	Property	Value
	Type	Variable: Reference
	Origin	AUTOSAR_ECUC
	Ref	Reference to node: EcucPartition

3.2.3 GptConfigurationOfOptApiServices

Container	GptConfigurationOfOptApiServices	
Description	This container contains all configuration switches for configuring optional API services of the GPT driver.	
Screenshot		
Properties	N/A	

3.2.3.1 GptDeinitApi

Variable	GptDeinitApi	
Description	Add/Remove the service Gpt_Deinit() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

3.2.3.2 GptTimeElapsedApi

Variable	GptTimeElapsedApi	
Description	Add/Remove the service Gpt_GetTimeElapsed() from the code.	
Screenshot	GptTimeElapsedApi  ☒ 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

3.2.3.3 GptTimeRemainingApi

Variable	GptTimeRemainingApi	
Description	Add/Remove the service Gpt_GetTimeRemaining() from the code.	
Screenshot	GptTimeRemainingApi  ☒ 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

3.2.3.4 GptVersionInfoApi

Variable	GptVersionInfoApi	
Description	Add/Remove the service Gpt_GetVersionInfo() from the code.	
Screenshot	GptVersionInfoApi  ☒ 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

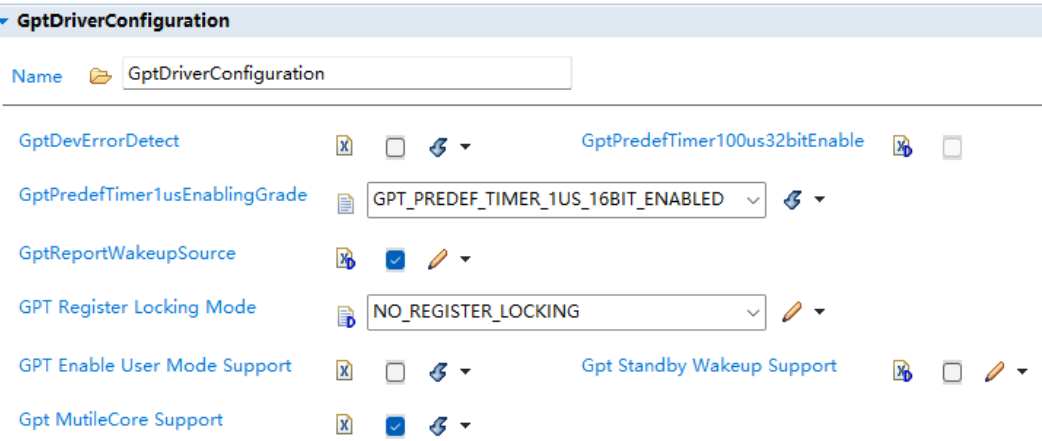
3.2.3.5 GptWakeupFunctionalityApi

Variable	GptWakeupFunctionalityApi	
Description	Add/Remove the services Gpt_SetMode(), Gpt_EnableWakeup() Gpt_DisableWakeup() and Gpt_Cbk_CheckWakeup() from the code.	
Screenshot	GptWakeupFunctionalityApi  ☐	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	false
	Editable	false

3.2.3.6 GptChangeNextTimeoutValueApi

Variable	GptChangeNextTimeoutValueApi	
Description	Enable settings for changing the channel counter compare value of a running counter. This API can be used FTU channels.	
Screenshot	GptChangeNextTimeoutValueApi  ☒ 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.4 GptDriverConfiguration

Container	GptDriverConfiguration	
Description	This container contains the module-wide configuration (parameters) of the GPT Driver.	
Screenshot		
Properties	N/A	

3.2.4.1 GptDevErrorDetect

Variable	GptDevErrorDetect	
Description	Enable/Disable development error detection.	
Screenshot	GptDevErrorDetect  ☒ 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

3.2.4.2 GptPredefTimer100us32bitEnable

Variable	GptPredefTimer100us32bitEnable	
Description	Enable/Disable the feature 100us/tick	
Screenshot	GptPredefTimer100us32bitEnable  ☐	
Properties	Property	Value

	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	false
	Readonly	true

3.2.4.3 GptPredefTimer1usEnablingGrade

Variable	GptPredefTimer1usEnablingGrade	
Description	Specify the grade of enabling the GPT Predef Timers with 1us tick duration.	
Screenshot	GptPredefTimer1usEnablingGrade  GPT_PREDEF_TIMER_1US_16BIT_ENABLED	
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	GPT_PREDEF_TIMER_1US_DISABLED
	Range	GPT_PREDEF_TIMER_1US_16BIT_ENABLED, GPT_PREDEF_TIMER_1US_16_24BIT_ENABLED, GPT_PREDEF_TIMER_1US_16_24_32BIT_ENABLED, GPT_PREDEF_TIMER_1US_DISABLED

3.2.4.4 GptReportWakeupSource

Variable	GptReportWakeupSource	
Description	Enable/Disable wakeup source reporting.	
Screenshot	GptReportWakeupSource  ☒ 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

3.2.4.5 GptRegisterLockingMode

Variable	GptRegisterLockingMode	
Description	N/A	
Screenshot	GPT Register Locking Mode  NO_REGISTER_LOCKING	
Properties	Property	Value
	Type	Variable: Enumeration
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	NO_REGISTER_LOCKING
	Range	NO_REGISTER_LOCKING, USER_MODE_SOFT_LOCKING, SUPERVISOR_MODE_SOFT_LOCKING

3.2.4.6 GptEnableUserModeSupport

Variable	GptEnableUserModeSupport	
Description	N/A	
Screenshot	GPT Enable User Mode Support    	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.4.7 GptStandbyWakeupSupport

Variable	GptStandbyWakeupSupport	
Description	The driver shall NOT CLEAR the interrupt flag, the interrupt enable bit and also should not disable the counter, during init (Gpt_Rtc_Init()) the flag is already set	
Screenshot	Gpt Standby Wakeup Support    	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

3.2.4.8 GptMultiCoreSupport

Variable	GptMultiCoreSupport	
Description	True: Gpt module support MultiCore. False :Gpt module doesn't support MultiCore.	
Screenshot	Gpt MultiCore Support     	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

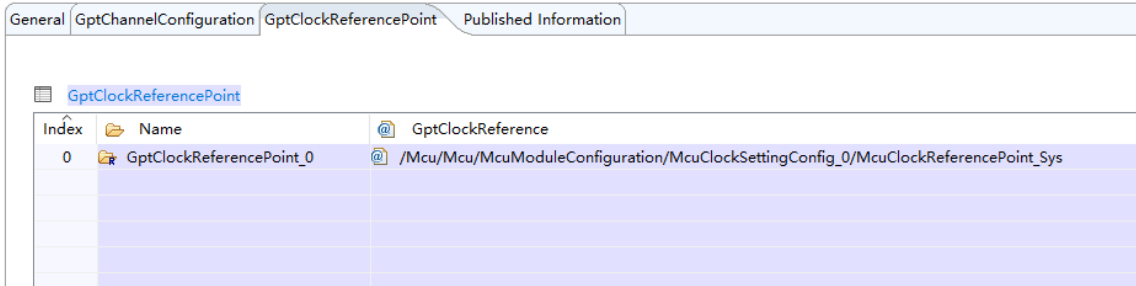
3.2.4.9 GptFtuGtbSupport

Variable	GptFtuGtbSupport	
Description	The FTU supports a global time base through the Global Time Base (GTB) feature.If enable,the ftu CNT won't start before calling Gpt_StartGlobalTime().	
Screenshot	Gpt Ftu Gtb Support     	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	false

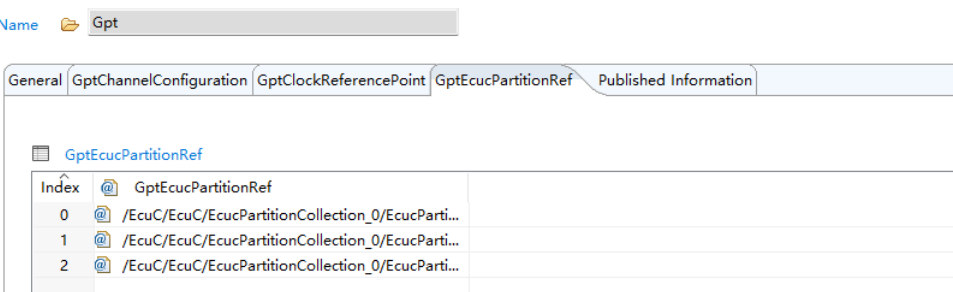
3.2.4.10 GptKernelEcucPartitionRef

Variable	GptKernelEcucPartitionRef	
Description	Maps the Gpt kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the Pwm driver is mapped to. This feature is not supported and is rejected.	
Screenshot		
Properties	Property	Value
	Type	Variable: Reference
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Ref	Reference to node: EcucPartition

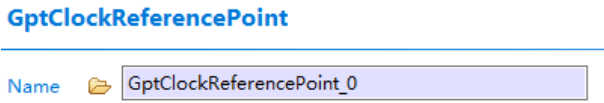
3.2.4.11 GptClockReferencePoint

Container	GptClockReferencePoint	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	List
	Min	1

3.2.4.12 GptEcuPartitionRef

Container	GptEcucPartitionRef	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	List
	Min	1
	Max	3

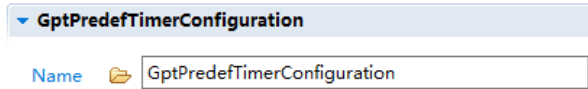
3.2.4.13 GptClockReferencePoint

Container	GptClockReferencePoint
Description	This container contains a parameter, which represents a reference to a container of the type McuClockReferencePoint (defined in module MCU).
Screenshot	 The screenshot shows the configuration for GptClockReferencePoint. It includes a 'Name' field with a folder icon and the text 'GptClockReferencePoint_0'.
Properties	N/A

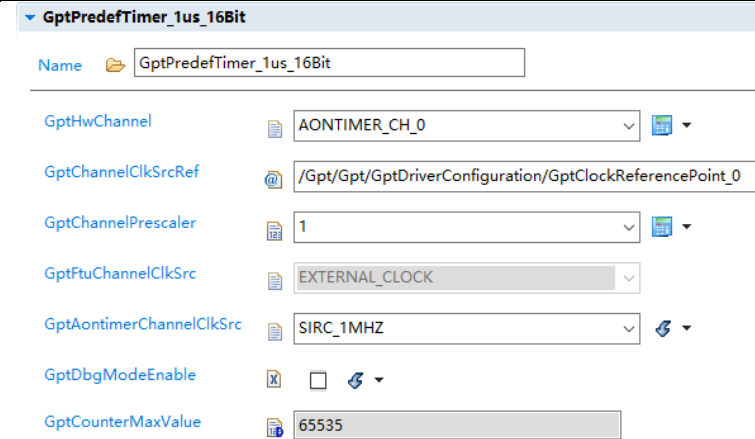
3.2.4.14 GptClockReference

Variable	GptClockReference	
Description	N/A	
Screenshot	GptClockReference  /Mcu/Mcu/McuModuleConfiguration/McuClockSettingConfig_0/McuClockReferencePoint_Sys	
Properties	Property	Value
	Type	Variable: Reference
	Origin	AUTOSAR_ECUC
	Ref	Reference to node: McuClockReferencePoint

3.2.5 GptPredefTimerConfiguration

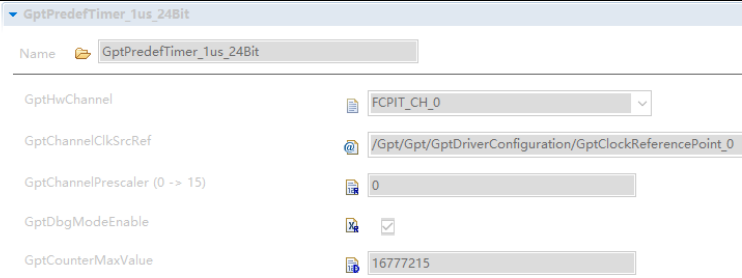
Container	GptPredefTimerConfiguration
Description	Container for configuring the Predefined Timer functionality.
Screenshot	 The screenshot shows the configuration for GptPredefTimerConfiguration. It includes a 'Name' field with a folder icon and the text 'GptPredefTimerConfiguration'.
Properties	N/A

3.2.5.1 GptPredefTimer_1us_16Bit

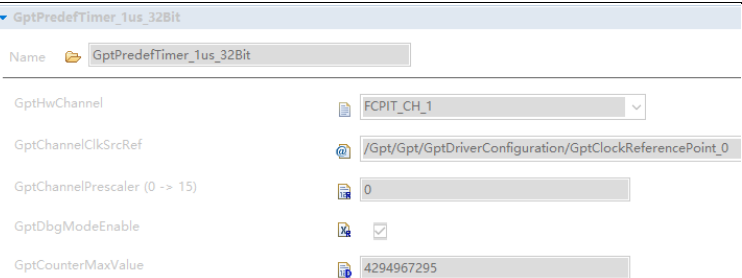
Container	GptPredefTimer_1us_16Bit
Description	This container contains the 1μs 16bit predef timer configuration (parameters) of the GPT Driver.
Screenshot	 The screenshot shows the configuration for GptPredefTimer_1us_16Bit. It includes a 'Name' field with a folder icon and the text 'GptPredefTimer_1us_16Bit'. Below the name field, there are several configuration parameters: - GptHwChannel: AONTIMER_CH_0 - GptChannelClkSrcRef: /Gpt/Gpt/GptDriverConfiguration/GptClockReferencePoint_0 - GptChannelPrescaler: 1 - GptFtuChannelClkSrc: EXTERNAL_CLOCK - GptAontimerChannelClkSrc: SIRC_1MHZ - GptDbgModeEnable: ☐ (disabled) - GptCounterMaxValue: 65535
Properties	N/A

3.2.5.2 GptPredefTimer_1us_24Bit

Container	GptPredefTimer_1us_24Bit
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Description	This container contains the 1μs 24bit predef timer configuration (parameters) of the GPT Driver.
Screenshot	
Properties	N/A

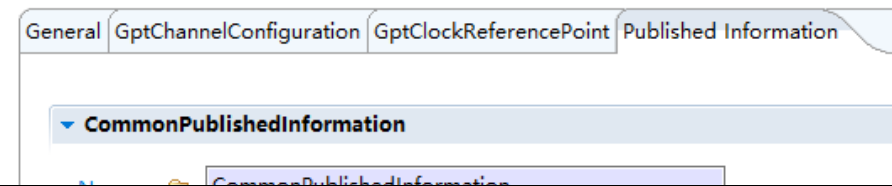
3.2.5.3 GptPredefTimer_1us_32Bit

Container	GptPredefTimer_1us_32Bit
Description	This container contains the 1μs 32bit predef timer configuration (parameters) of the GPT Driver.
Screenshot	
Properties	N/A

3.2.5.4 GptPredefTimer_100us_32Bit

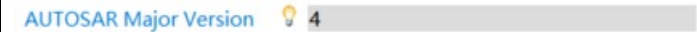
Container	GptPredefTimer_100us_32Bit	
Description	This container contains the 100μs 32bit predef timer configuration (parameters) of the GPT Driver.	
Screenshot	N/A	
Properties	Property	Value
	Type	Container
	Enable	false
	Visible	false

3.2.6 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	N/A	

3.2.6.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion
-----------------	-----------------------

Description	Major version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	4

3.2.6.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	6

3.2.6.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.6.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.6.5 SwMinorVersion

Variable	SwMinorVersion	
-----------------	----------------	--

Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	3

3.2.6.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.6.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	100

3.2.6.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	174

Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

- 1) In the GPT module, the FCPIT peripheral is only supported on FC7240.
- 2) If the GPT module uses the same FTU instance as the PWM/ICU/OCU module, the configuration tool will report an error.

Pwm

Name

General | PwmChannel | **PwmFtuModule** | PwmPowerStateConfig | Published Information

PwmFtuModule

Index	Name	Ftu Hard...	Ftu Mod...	Ftu Mod...	Ftu Mod...	Ftu Mod...	Reload F...	Default ...	Default ...
0	PwmFtuM...	FTU_0	PRESC_1	PWM_SYS...	/Mcu/Mc...	PWM_ED...	LDFQ_EAC...	10000	12000
1	PwmFtuM...	FTU_1	PRESC_1	PWM_SYS...	/Mcu/Mc...	PWM_ED...	LDFQ_EAC...	10000	12000

GptChannelConfiguration

Name

General

GptChannelId (0 -> 4294967295)

GptHwChannel

GptChannelMode

GptChannelTickFrequency (0 -> 150000000)

GptChannelClkSrcRef

GptChannelTickValueMax (65535 -> 4294967295)

GptDbgModeEnable ☒

GptIsrEnable ☒

GptNotification

GptFtuPrescaler

☒ GptEnableWakeup

- 3) If the GPT channel uses FTU and the clock source is from PCC, the GptFtuChannelClkSrc must select GPT_FTU_BUS_CLK.

GptChannelConfiguration

Name

GptChannel_WDG

General

GptChannelEcucPartitionRef

GptChannelId (0 -> 4294967295)

0

GptHwChannel

FTU_4_CH_0

GptChannelMode

GPT_CH_MODE_CONTINUOUS

GptChannelTickFrequency (0 -> 150000000)

750000.0

GptChannelClkSrcRef

/Gpt/Gpt/GptDriverConfiguration/GptClockReferencePoint_Ftu4

GptChannelTickValueMax (0 -> 4294967295)

65535

GptDbgModeEnable

☐
☒

GptEnableWakeup

☐

GptIsrEnable

☒

GptNotification

Wdg_Cbk_GptNotification0

Name

GptChannelConfiguration_FTU

GptFtuPrescaler

32

GptFtuChannelClkSrc

GPT_FTU_PCC_CLK

GptFtuGtb

☐
☒

GptFtuGtbRef

4.2 GPT Usage Common Steps

Basically, the GPT module can be configured by following the below 4 steps:

- 1) Configure the GPT channel clock reference point in GptClockReferencePoint tab.
- 2) Configure the GPT channel and link the GPT clock reference point in GptChannelConfiguration tab.
- 3) Choose which API will be used.
- 4) Generate configuration files.

4.3 GPT Channel Use FTU Demo

- ### 1) Configure GPT channel

In this demo, the GPT channel uses an FTU peripheral.

- a) GPT channel mode configured to continues mode, so the GptChannelMode selects GPT_CH_MODE_CONTINUES.
- b) FTU clock source from PCC and the external pin clock source is not used, so the GptFtuChannelClkSrc select GPT_FTU_PCC_CLK, and clock reference point select the specific point.
- c) GPT channel notification open, then enable GptIsrEnable first and write the notification function name, the name must not be NULL or NULL_PTR.
- d) Select the FTU prescaler and then the GptChannelTickFrequency will calculate automatically.

GptChannelConfiguration

Name

General GptChannelEcucPartitionRef

GptChannelId (0 -> 4294967295)

GptHwChannel

GptChannelMode

GptChannelTickFrequency (0 -> 15000000)

GptChannelClkSrcRef

GptChannelTickValueMax (0 -> 4294967295)

GptDbgModeEnable ☐

GptIsEnable ☒

GptNotification

Name

GptFtuPrescaler

GptFtuChannelClkSrc

GptFtuGtb ☐

GptFtuGtbRef

2) Configure Clock reference point

The clock reference point is a reference from the MCU module, so configure the MCU module first and create the reference point which GPT channel used.

Here the GPT channel uses an FTU peripheral, then we create a clock reference point item and link to this.

McuClockSettingConfig

Name

General FOSC SOSC SIRC SIRC32K FIRC PLL0 PLL1 System Clock PCC Clock Reference Point CSC0 SCM

McuClockReferencePoint

Ind...	Name	Mcu Clock Frequency Sel...	Mcu C...
0	McuClockReferencePoint_0	SYS_CORE_CLK	2.4E8
1	McuClockReferencePoint_Can0	FLEXCAN0_CLK	2.4E7
2	McuClockReferencePoint_FCSPI	FCSPi0_CLK	1.2E7
3	McuClockReferencePoint_Fcuart4	FCUART4_CLK	1.2E7
4	McuClockReferencePoint_Fcuart5	FCUART5_CLK	1.2E7
5	McuClockReferencePoint_Bus	SYS_BUS_CLK	1.2E8
6	McuClockReferencePoint_FCIIC1	FCIIC1_CLK	1.2E7
7	McuClockReferencePoint_Ftu4	FTU4_CLK	2.4E7
8	McuClockReferencePoint_Fosc	FOSC_CLK	2.4E7
9	McuClockReferencePoint_sent0	PLL1DIVL_CLK	7.5E7
10	McuClockReferencePoint_MSC	MSC0_CLK	600000...

Gpt

Name  Gpt

General GptChannelConfiguration GptClockReferencePoint GptEcucPartitionRef Gpt FtU Gtb reference Published Information

Bus clock reference point  /Mcu/Mcu/McuModuleConfiguration/MCU_Demo_FOSC24M/McuClockReferencePoint_Bus

 GptClockReferencePoint

Ind...	Name	GptClockReference
0	GptClockReferencePoint_Ftu4	/Mcu/Mcu/McuModuleConfiguration/MCU_Demo_FOSC24M/McuClockReferencePoint_Ftu4

3) Configure general configurations

Select which API will be used.

Gpt

Name  Gpt

General GptChannelConfiguration GptClockReferencePoint GptEcucPartitionRef Gpt FtU Gtb reference Published Information

Config Variant  VariantPostBuild 

GptConfigurationOfOptApiServices

Name  GptConfigurationOfOptApiServices

GptDeinitApi  ☒ 	GptEnableDisableNotificationApi  ☒ 
GptTimeElapsedApi  ☒ 	GptTimeRemainingApi  ☒ 
GptVersionInfoApi  ☒ 	GptWakeupFunctionalityApi  ☐ 
GptChangeNextTimeoutValueApi  ☐ 	

GptDriverConfiguration

Name  GptDriverConfiguration

GptDevErrorDetect  ☐ 	GptPredefTimer100us32bitEnable  ☐
GptPredefTimer1usEnablingGrade  PT_PREDEF_TIMER_1US_DISABLED 	
GptReportWakeupSource  ☒ 	
GPT Register Locking Mode  NO_REGISTER_LOCKING 	
GPT Enable User Mode Support  ☐ 	Gpt Standby Wakeup Support  ☐ 
Gpt MultiCore Support  ☐	Gpt FtU Gtb Support  ☒ 
 GptKernelEcucPartitionRef 	

4) Right click the module and select Generate Configuration to generate code after configuration is finished.